

Design a containerised 3

Class energy_storage	Feasibility conditional	BoM Readiness Ready	Score 9.0
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Executive Summary

Metric	Value	Unit	Confidence	Notes
Annual MWh delivered	1277.5	MWh/year	medium	Usable energy cycled once per day, adjusted for system efficiency.
Estimated CAPEX	41106	GBP	low	Summed only from admitted source-backed BoM lines.
Estimated annual OPEX	1644	GBP/year	low	Early estimate set at 4% of sourced CAPEX until supplier/service data is available.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class energy_storage (high); requirements 3; numeric facts 3; requirement trace 3/3 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	127 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	3 quantified requirement(s), 3 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected energy_storage with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	energy_storage has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	3 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class energy_storage (high); 3 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s): 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class energy_storage; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 11; submodules 46; components 204.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 3/3.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_held	pass	1	204 admitted sourcing evidence row(s), 1020 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 204; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class energy_storage; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage passed.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	204 admitted source evidence row(s), 204 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 438; source refs 612; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark no (prototype internal depth target).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class energy_storage; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	1465 provenance ref(s); 438 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	204 candidate BoM line(s); 204 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	612 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: prototype internal depth target; chain benchmark used: no; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Benchmark does not use chain-v2 input.

Architecture Freeze Gate

architecture_freeze_review_required

4/6 freeze areas pass; 2 review, 0 blocked. Modules/submodules/components 11/46/204. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces accepted_interfaces; requirements accepted_evidence; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.82	Component allocation allocation_ready; module engineering module_engineering_review_required; 1/11 modules pass; 0 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	pass	1	Interface verification accepted_interfaces; 3/3 required interfaces have carriers; 3 accepted.	none	Required interfaces have accepted reviewer evidence.
requirement_coverage	pass	1	Requirement coverage accepted_evidence; 3/3 requirements structurally ready or accepted; accepted ratio 1.	none	Every parsed requirement has accepted reviewer evidence.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 1; assumptions needing review 9.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	204 candidate BoM lines; 204 admitted sourcing evidence rows; 0 provenance violation(s); 0 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

2 closure row(s): 2 ready, 0 blocked. Queues S/E/V/R 1/1/0/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_struct ure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

Component Candidate Gate

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 204; critical 6/6; worklist rows 0; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	204 candidate BoM line(s), 204 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	204/204 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	204/204 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	6/6 class-critical part(s) are represented by candidate lines; 6 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	0/0 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	204 source evidence row(s), 1020 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorized

6/6 sourcing-authorization areas pass; 0 review, 0 blocked. Critical intake 0/0; full intake 0/0; admitted evidence 204; rejected evidence 0; provenance violations 0; authorized yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 204 candidate line(s), 0 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	0/0 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	0/0 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	204 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	1020 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

bom_admission_protocol_only

Display mode protocol_priced_fixture. 5/7 BoM-admission areas pass; 2 review, 0 blocked. Critical priced 6/6; source-backed claims 1020; provenance violations 0. Candidate BoM yes; priced BoM yes; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	pass	1	sourcing_authorized; authorized yes; 6/6 areas pass.	none	Sourcing is authorized, so the BoM may be shown as a candidate sourcing queue.
candidate_bom_scope	pass	1	204 candidate BoM line(s), 6 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	pass	1	204 admitted source row(s), 0 rejected source row(s), 204/204 priced line(s).	none	Admitted source rows may populate only their matched BoM fields.
critical_bom_admission	pass	1	6/6 critical line(s) have admitted unit costs; 0 critical line(s) remain unpriced.	none	Critical BoM lines are admitted with source-backed pricing.
source_reference_quality	review	0	protocol_source_only; 0/204 source-quality rows pass; 204 protocol fixture row(s).	none	Replace weak, protocol or placeholder source references before production BoM admission.
provenance_boundary	pass	1	1020 source-backed claim(s), 0 critical missing-source claim(s), 0 provenance violation(s).	none	BoM claims remain behind explicit source evidence.
display_mode	review	0.5	Display mode will be protocol_priced_fixture.	none	Show protocol-priced rows only as fixture evidence; do not use for production cost claims.

Report Readiness Gate

publishable

6/6 sections at or above 8. Mean score 9. Architecture ready: yes. Critical unpriced lines: 0. Verification accepted: 25/25; unaccepted: 0. Verification acceptance ratio: 1.

No deterministic promotion blockers.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 11/46/204. Calculations within envelope 2/3; calculation blockers 0. Assumption review rows 9; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	11 modules, 46 submodules and 204 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	46/46 submodules ready; 6/6 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	3/3 required interfaces have endpoint carriers; 3 accepted with evidence.	none	Required interfaces have accepted reviewer evidence.
requirement_coverage	pass	1	3/3 requirements are structurally ready or accepted; accepted evidence ratio 1.	none	Requirements have accepted reviewer evidence.
calculation_envelope	review	0.67	2/3 calculations are within envelope; 1 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.57	12/21 assumptions are brief-supported or model-present; 9 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_protocol_only

5/7 costing areas pass; 2 review, 0 blocked. Critical priced lines 6/6; unpriced critical lines 0. Source-backed claims 1020; provenance violations 0; sourcing evidence 0/204 production-ready, 204 protocol. Source quality: protocol_source_only (0/204 pass, 0 placeholder URL, 0 quote-anchored). BoM total £41,106; CAPEX £41,106.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	pass	1	204/204 BoM lines have admitted pricing; rejected source records 0.	none	Sourcing admission has priced every BoM line.
critical_line_pricing	pass	1	6/6 critical lines have admitted unit costs.	none	Critical BoM lines have admitted pricing.
source_backed_provenance	pass	1	1020 source-backed claims; 0 critical missing-source claims; 0 provenance violations.	none	BoM claim provenance is attached for admitted claims.
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line-level costing.
sourcing_evidence_authenticity	review	0	0/204 sourcing evidence rows are production-ready; 204 protocol fixture, 0 review/missing metadata.	sourcing:1:lfprismatic_cells: source evidence is a protocol fixture and proves mechanics only. sourcing:2:cell_to_cell_busbar: source evidence is a protocol fixture and proves mechanics only. sourcing:3:cell_terminal_hardware_set: source evidence is a protocol fixture and proves mechanics only. sourcing:4:cell_insulation_pad: source evidence is a protocol fixture and proves mechanics only. sourcing:5:module_fuse: source evidence is a protocol fixture and proves mechanics only. plus 199 more	Replace protocol or weak sourcing evidence with retrievable supplier/catalogue evidence and complete metadata.
source_reference_quality	review	0	0/204 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 204 fresh timestamp.	none	Replace protocol fixture with a retrievable supplier/catalogue URL before production costing.
cost_totals	pass	1	BoM total 41106 GBP; CAPEX 41106 GBP; annual OPEX 1644 GBP.	none	Cost totals are internally populated from admitted BoM pricing.

Evidence Acquisition Plan

No missing sourcing or reviewer evidence acquisition rows remain.

Document Trust Gate

architecture_review_only

6/7 trust areas pass; 1 review, 0 blocked. Architecture ready: yes. Claim gate: claim_evidence_complete. Reviewer evidence 25/25. Source-backed BoM claims 1020; critical missing BoM claims 0; provenance violations 0. Evidence authenticity: protocol_only (0/229 production-ready, 229 protocol).

No document-trust blockers.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	11 modules, 46 submodules and 3 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	pass	1	6/6 scored sections are at or above 8; mean 9.	none	Keep score ledger attached; external evidence gates may still block trust.
claim_evidence	pass	1	1295/1295 claims pass; 0 blocked/source-required, 0 review-required.	none	No claim-evidence action required.
requirement_assurance	pass	1	3/3 requirements have accepted assurance; 0 need review, 0 blocked, 0 unlinked.	none	All requirement assurance rows have accepted evidence.
reviewer_evidence	pass	1	25/25 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 0 pending/blocked.	none	All non-sourcing verification activities have accepted reviewer evidence.
bom_provenance	pass	1	1020 source-backed BoM claims, 0 critical missing-source claims, 0 provenance violations.	none	Keep source refs attached and refresh before procurement use.
evidence_authenticity	review	0	0/229 evidence row(s) are production-ready; 229 protocol fixture, 0 review/missing metadata.	sourcing:1:lfprismatic_cells: source evidence is a protocol fixture and proves mechanics only. sourcing:2:cell_to_cell_busbar: source evidence is a protocol fixture and proves mechanics only. sourcing:3:cell_terminal_hardware_set: source evidence is a protocol fixture and proves mechanics only. sourcing:4:cell_insulation_pad: source evidence is a protocol fixture and proves mechanics only. sourcing:5:module_fuse: source	Replace protocol fixtures or weak references with production evidence before calling the document trusted.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				evidence is a protocol fixture and proves mechanics only. plus 224 more	

Evidence Authenticity Gate

protocol_only

0/229 evidence rows are production-ready. Protocol fixtures 229; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
sourcing	lfp_prismatic_cells	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	cell_to_cell_busbar	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	cell_terminal_hardware_set	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/cell_terminal_hardware_set	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	cell_insulation_pad	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/cell_insulation_pad	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	module_fuse	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/module_fuse	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	module_steel_frame	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/module_steel_frame	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	compression_plate	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/c	source evidence is a protocol fixture and	Replace protocol or non-external BoM evidence with a

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
				ompression_plate	proves mechanics only.	retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	compression_tie_rod_set	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/compression_tie_rod_set	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	module_top_cover	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/module_top_cover	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	rack_label_plate	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/rack_label_plate	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	bms_slave_pcb	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/bms_slave_pcb	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	afe_monitor_ic	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/afe_monitor_ic	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	ntc_thermistor_harness	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/ntc_thermistor_harness	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
						refreshed timestamp.
sourcing	isolated_can_transceiver	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/isolated_can_transceiver	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	conformal_coating	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/conformal_coating	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	dc_contactor_or_breaker	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/dc_contactor_or_breaker	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	precharge_resistor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/precharge_resistor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	hv_fuse_holder	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/hv_fuse_holder	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	pack_current_sensor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/pack_current_sensor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
sourcing	service_disconnect	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/service_disconnect	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	cell_voltage_sense_loom	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/cell_voltage_sense_loom	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	rack_can_drop_cable	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/rack_can_drop_cable	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	harness_strain_relief_rail	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/harness_strain_relief_rail	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	low_smoke_cable_sleeve	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/low_smoke_cable_sleeve	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	rack_cold_plate_manifold	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/rack_cold_plate_manifold	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	thermally_conductive_gap_pad	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/t	source evidence is a protocol fixture and	Replace protocol or non-external BoM evidence with a

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
				hermally_conduc tive_gap_pad	proves mechanics only.	retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	coolant_quick_c onnector_pair	protocol_fixture	protocol_fixt ure	test- fixture://review- evidence- closure/source/c oolant_quick_co nnector_pair	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	rack_drip_tray	protocol_fixture	protocol_fixt ure	test- fixture://review- evidence- closure/source/r ack_drip_tray	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	leak_detection_r ope	protocol_fixture	protocol_fixt ure	test- fixture://review- evidence- closure/source/l eak_detection_r ope	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	pcs_inverter	protocol_fixture	protocol_fixt ure	test- fixture://review- evidence- closure/source/p cs_inverter	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	igbt_power_mod ule	protocol_fixture	protocol_fixt ure	test- fixture://review- evidence- closure/source/i gbt_power_mod ule	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	dc_link_capacito r_bank	protocol_fixture	protocol_fixt ure	test- fixture://review- evidence- closure/source/d c_link_capacitor _bank	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
						refreshed timestamp.
sourcing	gate_driver_board	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/gate_driver_board	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	inverter_control_board	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/inverter_control_board	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	liquid_cooled_heatsink	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/liquid_cooled_heatsink	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	cast_resin_transformer	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/cast_resin_transformer	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	hv_winding	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/hv_winding	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	lv_winding	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/lv_winding	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
sourcing	temperature_probe	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/temperature_probe	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	anti_vibration_mount	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/anti_vibration_mount	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	grid_filter_choke	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/grid_filter_choke	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	ac_filter_capacitor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/ac_filter_capacitor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	emi_filter_assembly	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/emi_filter_assembly	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	protective_earth_bar	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/protective_earth_bar	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	grid_protection_relay	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/g	source evidence is a protocol fixture and	Replace protocol or non-external BoM evidence with a

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
				rid_protection_r elay	proves mechanics only.	retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	voltage_sensing _fuse_set	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/v oltage_sensing_ fuse_set	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	relay_test_block	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/r elay_test_block	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	trip_circuit_moni tor	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/tr ip_circuit_monit or	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	g99_settings_la bel	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/g 99_settings_lab el	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	corten_steel_sid e_panel	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/c orten_steel_side _panel	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	corner_casting	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/c orner_casting	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
						refreshed timestamp.
sourcing	marine_plywood_floor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/marine_plywood_floor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	roof_panel	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/roof_panel	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	steel_stud_frame	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/steel_stud_frame	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	fire_rated_board	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/fire_rated_board	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	mineral_wool_insulation	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/mineral_wool_insulation	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	intumescent_sealant	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/intumescent_sealant	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
sourcing	floor_reinforcement_plate	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/floor_reinforcement_plate	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	rack_mounting_rail	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/rack_mounting_rail	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	vibration_isolator	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/vibration_isolator	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	anchor_bolt_set	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/anchor_bolt_set	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	gland_plate_assembly	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/gland_plate_assembly	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	firestop_collar_set	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/firestop_collar_set	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	emc_cable_gland	protocol_fixture	protocol_fixture	test-fixture://review-evidence-	source evidence is a protocol fixture and	Replace protocol or non-external BoM evidence with a

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
				closure/source/e mc_cable_gland	proves mechanics only.	retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	rain_drip_shield	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/r ain_drip_shield	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	penetration_sche dule_label	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/p enetration_sche dule_label	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	dc_voltage_tran sducer	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/d c_voltage_trans ducer	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	hall_current_sen sor	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/h all_current_sens or	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	insulation_monit oring_device	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/i nsulation_monit oring_device	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and refreshed timestamp.
sourcing	auxiliary_meterin g_transducer	protocol_fixture	protocol_fixture	test- fixture://review- evidence- closure/source/a uxiliary_meterin g_transducer	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalog ue URL and

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
						refreshed timestamp.
sourcing	temperature_humidity_sensor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/temperature_humidity_sensor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	hydrogen_detector	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/hydrogen_detector	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	smoke_aspirating_detector	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/smoke_aspirating_detector	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	door_position_switch	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/door_position_switch	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	coolant_inlet_thermistor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/coolant_inlet_thermistor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	coolant_outlet_thermistor	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/coolant_outlet_thermistor	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
sourcing	flow_meter	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/flow_meter	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	pressure_transducer	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/pressure_transducer	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	fault_recorder_module	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/fault_recorder_module	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.
sourcing	event_timestamp_logger	protocol_fixture	protocol_fixture	test-fixture://review-evidence-closure/source/event_timestamp_logger	source evidence is a protocol fixture and proves mechanics only.	Replace protocol or non-external BoM evidence with a retrievable supplier/catalogue URL and refreshed timestamp.

Showing first 80 of 229 evidence-authenticity rows; export CSV for the full ledger.

Evidence Replacement Plan

229 replacement row(s): 204 sourcing, 25 verification. Protocol 229; review-required 0; missing metadata 0. Blocks BoM 204; blocks publish 229. Next row: replace:1:sourcing:1:lpf_prismatic_cells.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
1	sourcing	lfp_prismatic_cells	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:1:lfp_prismatic_cells with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
2	sourcing	cell_to_cell_busbar	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:2:cell_to_cell_busbar with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
3	sourcing	cell_terminal_hardware_set	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:3:cell_terminal_hardware_set with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
4	sourcing	cell_insulation_pad	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:4:cell_insulation_pad with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
5	sourcing	module_fuse	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:5:module_fuse with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
6	sourcing	module_steel_frame	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:6:module_steel_frame with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
7	sourcing	compression_plate	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:7:compression_plate with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
8	sourcing	compression_tie_rod_set	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:8:compression_tie_rod_set with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
9	sourcing	module_top_cover	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:9:module_top_cover with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
10	sourcing	rack_label_plate	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:10: rack_label_plate with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
11	sourcing	bms_slave_pcb	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:11: bms_slave_pcb with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
12	sourcing	afe_monitor_ic	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:12: afe_monitor_ic with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
13	sourcing	ntc_thermistors_harness	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:13: ntc_thermistors_harness with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
14	sourcing	isolated_cann_transceiver	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:14: isolated_cann_transceiver with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
15	sourcing	conformal_coating	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:15: conformal_coating with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
16	sourcing	dc_contactor_or_breaker	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:16: dc_contactor_or_breaker with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
17	sourcing	precharge_resistor	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:17: precharge_resistor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
18	sourcing	hv_fuse_holder	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:18: hv_fuse_holder with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
19	sourcing	pack_current_sensor	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:19: pack_current_sensor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
20	sourcing	service_disconnect	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:20:service_disconnect with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
21	sourcing	cell_voltage_sense_loop	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:21: cell_voltage_sense_loop with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
22	sourcing	rack_can_drop_cable	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:22: rack_can_drop_cable with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
23	sourcing	harness_strain_relief_rail	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:23: harness_strain_relief_rail with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
24	sourcing	low_smoke_cable_sleeve	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:24: low_smoke_cable_sleeve with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
25	sourcing	rack_cold_plate_manifold	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:25: rack_cold_plate_manifold with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
26	sourcing	thermally_conductive_gap_pad	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:26: thermally_conductive_gap_pad with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
27	sourcing	coolant_quick_connector_pair	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:27: coolant_quick_connector_pair with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
28	sourcing	rack_drip_tray	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:28:rack_drip_tray with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
29	sourcing	leak_detection_rope	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:29:leak_detection_rope with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
30	sourcing	pcs_inverter	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:30:pcs_inverter with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
31	sourcing	igbt_power_module	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:31:igbt_power_module with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
32	sourcing	dc_link_capacitor_bank	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:32:dc_link_capacitor_bank with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
33	sourcing	gate_driver_board	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:33:gate_driver_board with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
34	sourcing	inverter_control_board	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:34:inverter_control_board with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
35	sourcing	liquid_cooled_heatsink	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:35:liquid_cooled_heatsink with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
36	sourcing	cast_resin_transformer	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:36:cast_resin_transformer with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
37	sourcing	hv_winding	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:37: hv_winding with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
38	sourcing	lv_winding	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:38:lv_winding with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
39	sourcing	temperature_probe	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:39:temperature_probe with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
40	sourcing	anti_vibration_mount	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:40:anti_vibration_mount with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
41	sourcing	grid_filter_choke	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:41:grid_filter_choke with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
42	sourcing	ac_filter_capacitor	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:42:ac_filter_capacitor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
43	sourcing	emi_filter_assembly	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:43:emi_filter_assembly with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
44	sourcing	protective_earth_bar	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:44:protective_earth_bar with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
45	sourcing	grid_protection_relay	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:45:grid_protection_relay with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
46	sourcing	voltage_sensing_fuse_set	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:46:voltage_sensing_fuse_set with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
47	sourcing	relay_test_block	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:47:relay_test_block with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
48	sourcing	trip_circuit_monitor	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:48:trip_circuit_monitor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
49	sourcing	g99_settings_label	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:49:g99_settings_label with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
50	sourcing	corten_steel_side_panel	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:50:corten_steel_side_panel with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
51	sourcing	corner_casting	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:51:corner_casting with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
52	sourcing	marine_ply wood_floor	protocol_fix ture	external_supplier_or_catalogue_url	external_url	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	yes	Replace sourcing:52:marine_ply wood_floor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
53	sourcing	roof_panel	protocol_fix ture	external_supplier_or_catalogue_url	external_url	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	yes	Replace sourcing:53:roof_panel with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
54	sourcing	steel_stud_frame	protocol_fix ture	external_supplier_or_catalogue_url	external_url	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	yes	Replace sourcing:54:steel_stud_frame with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
55	sourcing	fire_rated_board	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:55:fire_rated_board with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
56	sourcing	mineral_wool_insulation	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:56:mineral_wool_insulation with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
57	sourcing	intumescent_sealant	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:57:intumescent_sealant with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
58	sourcing	floor_reinforcement_plate	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:58:floor_reinforcement_plate with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
59	sourcing	rack_mounting_rail	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:59:rack_mounting_rail with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
60	sourcing	vibration_isolator	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:60:vibration_isolator with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
61	sourcing	anchor_bolt_set	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:61: anchor_bolt_set with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
62	sourcing	gland_plate_assembly	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:62:gland_plate_assembly with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
63	sourcing	firestop_collar_set	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:63:firestop_collar_set with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
64	sourcing	emc_cable_gland	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:64:emc_cable_gland with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
65	sourcing	rain_drip_shield	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:65:rain_drip_shield with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
66	sourcing	penetration_schedule_label	protocol_fix ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:66:penetration_schedule_label with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
67	sourcing	dc_voltage_transducer	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:67:dc_voltage_transducer with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
68	sourcing	hall_current_sensor	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:68:hall_current_sensor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
69	sourcing	insulation_monitoring_device	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:69:insulation_monitoring_device with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
70	sourcing	auxiliary_metering_transducer	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:70: auxiliary_metering_transducer with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
71	sourcing	temperature_humidity_sensor	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:71: temperature_humidity_sensor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
72	sourcing	hydrogen_detector	protocol_fixture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:72: hydrogen_detector with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
73	sourcing	smoke_aspirating_detector	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:73: smoke_aspirating_detector with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
74	sourcing	door_position_switch	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:74: door_position_switch with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
75	sourcing	coolant_inlet_thermistor	protocol_fixature	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:75: coolant_inlet_thermistor with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
76	sourcing	coolant_outlet_thermist or	protocol_fix ture	external_supplier_or_catalogue_url	external_url	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	yes	Replace sourcing:76: coolant_outlet_thermist or with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
77	sourcing	flow_meter	protocol_fix ture	external_supplier_or_catalogue_url	external_url	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	yes	Replace sourcing:77: flow_meter with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
78	sourcing	pressure_transducer	protocol_fix ture	external_supplier_or_catalogue_url	external_url	component WordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	yes	Replace sourcing:78: pressure_transducer with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

#	Kind	Subject	Current Status	Replacement Target	Accepted Classes	Required Fields	Blocks BoM	Action
79	sourcing	fault_recorder_module	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:79: fault_recorder_module with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.
80	sourcing	event_timestamp_logger	protocol_fix_ture	external_supplier_or_catalogue_url	external_url	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	yes	Replace sourcing:80 :event_timestamp_logger with source-backed supplier/catalogue evidence before this BoM line can support trusted publication.

Showing first 80 of 229 replacement rows; export CSV for the full worklist.

Trust Repair Plan

2 package(s): 2 ready, 0 waiting. Next package: evidence_authenticity_review. Closure rows 10; trust blockers 229. Trust verdict: architecture_review_only.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
3	Resolve Engineering Review Questions	ready	closure only	10	0	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumption.csv; *.engineering-calculations.csv	Requirements assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	229	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence rows remain.

Claim Ledger

1295 claims: 3 brief-supplied, 270 reviewer-accepted, 1022 source-backed, 0 calculated needing review, 0 generated needing review, 0 source-required, 0 blocked. Blocking: 0 BoM, 0 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
brief_requirement	brief_sulfilled	brief	Energy capacity	Energy capacity = 3.5 MWh.	brief:input.brief	no	no	Confirm requirement value, tolerance and acceptance condition with the project owner.
brief_requirement	brief_sulfilled	brief	Mass constraint	Mass constraint = 28000 kg.	brief:input.brief	no	no	Confirm requirement value, tolerance and acceptance condition with the project owner.
brief_requirement	brief_sulfilled	brief	Power rating	Power rating = 1 MW.	brief:input.brief	no	no	Confirm requirement value, tolerance and acceptance condition with the project owner.
bom_source_field	source_backed	source_evidence	BMS master controller / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/bms_master_controller	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	BMS master controller / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/bms_master_controller	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	BMS master controller / mpn	mpn = PROTOCOL-84-bms_master_controller.	test-fixture://review-evidence-closure/source/bms_master_controller	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	BMS master controller / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/bms_master_controller	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	BMS master controller / unit_cost_gbp	unit_cost_gbp = 183.	test-fixture://review-evidence-closure/source/bms_master_controller	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	contactor driver output / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/contactor_driver_output	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	contactor driver output / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/contactor_driver_output	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	contactor driver output / mpn	mpn = PROTOCOL-86-contactor_driver_output.	test-fixture://review-evidence-closure/source/contactor_driver_output	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	contactor driver output / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/contactor_driver_output	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	contactor driver output / unit_cost_gbp	unit_cost_gbp = 185.	test-fixture://review-evidence-closure/source/contactor_driver_output	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	isolation interface / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/isolation_interface	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	isolation interface / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/isolation_interface	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	isolation interface / mpn	mpn = PROTOCOL-85-isolation_interface.	test-fixture://review-evidence-closure/source/isolation_interface	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	isolation interface / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/isolation_interface	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	isolation interface / unit_costGBP	unit_costGBP = 184.	test-fixture://review-evidence-closure/source/isolation_interface	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SOC estimation firmware / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/soc_estimation_firmware	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SOC estimation firmware / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/soc_estimation_firmware	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	SOC estimation firmware / mpn	mpn = PROTOCOL-87-soc_estimation_firmware.	test-fixture://review-evidence-closure/source/soc_estimation_firmware	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SOC estimation firmware / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/soc_estimation_firmware	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SOC estimation firmware / unit_cost_gbp	unit_cost_gbp = 186.	test-fixture://review-evidence-closure/source/soc_estimation_firmware	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	cybersecurity firewall / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/cybersecurity_firewall	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	cybersecurity firewall / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/cybersecurity_firewall	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	cybersecurity firewall / mpn	mpn = PROTOCOL-95-cybersecurity_firewall.	test-fixture://review-evidence-closure/source/cybersecurity_firewall	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	cybersecurity firewall / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/cybersecurity_firewall	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	cybersecurity firewall / unit_cost_gbp	unit_cost_gbp = 194.	test-fixture://review-evidence-closure/source/cybersecurity_firewall	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	Modbus TCP gateway / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/modbus_tcp_gateway	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	Modbus TCP gateway / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/modbus_tcp_gateway	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	Modbus TCP gateway / mpn	mpn = PROTOCOL-92-modbus_tcp_gateway.	test-fixture://review-evidence-closure/source/modbus_tcp_gateway	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	Modbus TCP gateway / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/modbus_tcp_gateway	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	Modbus TCP gateway / unit_cost_gbp	unit_cost_gbp = 191.	test-fixture://review-evidence-closure/source/modbus_tcp_gateway	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SCADA protocol adapter / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/scada_protocol_adapter	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	SCADA protocol adapter / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/scada_protocol_adapter	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SCADA protocol adapter / mpn	mpn = PROTOCOL-93-scada_protocol_adapter.	test-fixture://review-evidence-closure/source/scada_protocol_adapter	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SCADA protocol adapter / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/scada_protocol_adapter	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	SCADA protocol adapter / unit_costGBP	unit_costGBP = 192.	test-fixture://review-evidence-closure/source/scada_protocol_adapter	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	time synchronisation module / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/time_synchronisation_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	time synchronisation module / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/time_synchronisation_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	time synchronisation module / mpn	mpn = PROTOCOL-94-time_synchronisation_module.	test-fixture://review-evidence-closure/source/time_synchronisation_module	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	time synchronisation module / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/time_synchronisation_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	time synchronisation module / unit_cost_gbp	unit_cost_gbp = 193.	test-fixture://review-evidence-closure/source/time_synchronisation_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	account key switch / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/account_key_switch	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	account key switch / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/account_key_switch	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	account key switch / mpn	mpn = PROTOCOL-98-account_key_switch.	test-fixture://review-evidence-closure/source/account_key_switch	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	account key switch / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/account_key_switch	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	account key switch / unit_cost_gbp	unit_cost_gbp = 197.	test-fixture://review-evidence-closure/source/account_key_switch	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	configuration backup module / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/configuration_backup_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	configuration backup module / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/configuration_backup_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	configuration backup module / mpn	mpn = PROTOCOL-100-configuration_backup_module.	test-fixture://review-evidence-closure/source/configuration_backup_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	configuration backup module / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/configuration_backup_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	configuration backup module / unit_costGBP	unit_costGBP = 199.	test-fixture://review-evidence-closure/source/configuration_backup_module	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	firmware escrow tag / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/firmware_escrow_tag	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	firmware escrow tag / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/firmware_escrow_tag	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	firmware escrow tag / mpn	mpn = PROTOCOL-99-firmware_escrow_tag.	test-fixture://review-evidence-closure/source/firmware_escrow_tag	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	firmware escrow tag / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/firmware_escrow_tag	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	firmware escrow tag / unit_cost_gbp	unit_cost_gbp = 198.	test-fixture://review-evidence-closure/source/firmware_escrow_tag	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	syslog collector / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/syslog_collector	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	syslog collector / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/syslog_collector	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	syslog collector / mpn	mpn = PROTOCOL-97-syslog_collector.	test-fixture://review-evidence-closure/source/syslog_collector	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	syslog collector / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/syslog_collector	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	syslog collector / unit_cost_gbp	unit_cost_gbp = 196.	test-fixture://review-evidence-closure/source/syslog_collector	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	VPN access appliance / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/vpn_access_appliance	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	VPN access appliance / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/vpn_access_appliance	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	VPN access appliance / mpn	mpn = PROTOCOL-96-vpn_access_appliance.	test-fixture://review-evidence-closure/source/vpn_access_appliance	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	VPN access appliance / supplier	supplier = Protocol Evidence Supplier.	test-fixture://review-evidence-closure/source/vpn_access_appliance	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	VPN access appliance / unit_cost_gbp	unit_cost_gbp = 195.	test-fixture://review-evidence-closure/source/vpn_access_appliance	no	no	Keep source record attached; refresh evidence before procurement use.
bom_source_field	source_backed	source_evidence	cellular router / lead_time_weeks	lead_time_weeks = 8.	test-fixture://review-evidence-closure/source/cellular_router	no	no	Keep source record attached; refresh evidence before procurement use.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_backed	source_evidence	cellular router / manufacturer	manufacturer = Protocol Evidence Manufacturer.	test-fixture://review-evidence-closure/source/cellular_router	no	no	Keep source record attached; refresh evidence before procurement use.

Showing first 70 of 1295 claims; export CSV for the full claim ledger.

Claim Evidence Gate

claim_evidence_complete

1295/1295 claims pass by being brief-supplied, source-backed or reviewer-accepted. 0 blocked/source-required claims, 0 review-required claims. Claim pass ratio 1. Blocking: 0 BoM, 0 publishable.

No claim-evidence promotion blockers.

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	3	3	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	complete	57	57	0	0	0	0	0	1	No claim-evidence action required for this area.
engineering_math	complete	6	6	0	0	0	0	0	1	No claim-evidence action required for this area.
bom_sourcing	complete	1224	1224	0	0	0	0	0	1	No claim-evidence action required for this area.
compliance_risk	complete	5	5	0	0	0	0	0	1	No claim-evidence action required for this area.

Evidence Gap Register

10 open evidence gaps: 0 blockers, 10 major, 0 minor. Closure paths: 0 sourcing intake, 0 verification intake, 10 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 0, publishable 10.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
calculation_review	major	needs review	BESS annual nameplate throughput	1277.5 MWh/year assumes one full equivalent cycle per day.	engineering_review	publishable	Confirm cycles per year, usable depth of discharge, round-trip efficiency and degradation assumptions.
assumption_review	major	needs review	Annual MWh delivered	Annual MWh delivered uses capacity_mwh * cycles_per_year * efficiency with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	major	needs review	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	engineering_review	publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	engineering_review	publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	BESS C-Rate	BESS C-Rate currently evaluates as pass against 0.1C to 1.0C for typical grid-storage duration systems.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Rated Duration	Rated Duration currently evaluates as pass against 1 to 6 hours for common containerised BESS applications.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Container-Level Energy Density	Container-Level Energy Density currently evaluates as pass against 50 to 200 Wh/kg at	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				container/system level before detailed structural mass review.			rationale for this envelope.
assumption_review	major	needs review	IEC-62619	Industrial lithium battery safety is relevant to this product class in Global / UK-adopted.	engineering_review	publishable	Cell certificates and pack-level safety validation.
assumption_review	major	needs review	UL-9540A	Thermal runaway fire propagation test is relevant to this product class in US / investor diligence.	engineering_review	publishable	Cell, module and unit-level propagation evidence.
assumption_review	major	needs review	G99	UK grid connection requirements is relevant to this product class in UK.	engineering_review	publishable	Protection settings, relay test certificates and commissioning records.

Closure Plan

1 active phases. Next phase: engineering_review. Rows by phase: architecture 0, sourcing 0, engineering 10, verification 0, score repair 0.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
3	Complete Engineering Review	10	0/10/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:c ompliance:G9 9; assumption:c ompliance:IEC -62619; assumption:c ompliance:UL -9540A; assumption:m etric:capex_g bp; assumption:m etric:headline _output; assumption:m etric:opex_an nual_gbp; assumption:sa nity:bess_c_r ate; assumption:sa nity:bess_con tainer_energy _density; plus 2 more

Depth Benchmark

prototype internal depth target. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

7 meeting benchmark, 0 below benchmark, average comparable ratio 1.1.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	11 modules	10 modules	1.1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	46 submodules	40 submodules	1.15	meets	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	204 components	200 components	1.02	meets	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	50 links	45 links	1.11	meets	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	1 ratio	0.8 ratio	1.25	meets	Prototype target expects most critical lines to become source-backed before publication.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	6 sections	6 sections	1	meets	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	25 activities	24 activities	1.04	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a containerised 3.5 MWh battery energy storage system with 1 MW PCS, 28 tonne gross mass limit, and LFP prismatic cells.

Requirement	Value	Unit
Energy capacity	3.5	MWh
Power rating	1	MW
Mass constraint	28000	kg

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Energy capacity	covered	3.5 MWh	Energy Storage Source / Cell string; Power Conversion System / PCS inverter bridge; DC And AC Power Distribution / Main DC busbars	metric:headline_output; sanity:bess_c_rate; sanity:bess_duration; sanity:bess_container_energy_density	Energy capacity is traced through storage, conversion, distribution and sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.
Power rating	covered	1 MW	Power Conversion System / PCS inverter bridge; DC And AC Power Distribution / Main DC busbars; Control Compute And Communication / BMS master controller	sanity:bess_c_rate; sanity:bess_duration	Power rating is traced through PCS, distribution and power-to-energy sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.
Mass constraint	covered	28000 kg	Container Structure And Fit-Out / ISO container shell; Maintenance And Serviceability / Access doors and hardware	sanity:bess_container_energy_density	Mass is traced through structure and system-level energy-density sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

accepted_evidence

3 requirement row(s): 3 accepted with evidence, 0 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 1.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Energy capacity	accepted	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity; accepted_verification	none	energy_storage_source; energy_conversion_transduction; power_distribution	design_review:energy_storage_source; calculation:energy_storage_source; design_review:energy_conversion_transduction; calculation:energy_conversion_transduction; design_review:power_distribution; calculation:power_distribution	Requirement has accepted architecture/reviewer evidence in the current ledger.
Mass constraint	accepted	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity; accepted_verification	none	structure_containment; maintenance_serviceability	design_review:structure_containment; calculation:structure_containment; design_review:maintenance_serviceability; calculation:maintenance_serviceability	Requirement has accepted architecture/reviewer evidence in the current ledger.
Power rating	accepted	architecture_link ; submodule_link; component_link; calculation_link; review_question; verification_activity; accepted_verification	none	energy_conversion_transduction; power_distribution; control_compute_communication	design_review:energy_conversion_transduction; calculation:energy_conversion_transduction; design_review:control_compute_communication; calculation:control_compute_communication; design_review:power_distribution; calculation:power_distribution	Requirement has accepted architecture/reviewer evidence in the current ledger.

Engineering Assurance Matrix

3 requirements: 3 accepted, 0 ready, 0 needing review, 0 blocked, 0 unlinked. 3 have calculation links, 3 have review questions, 3 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Energy capacity	accepted	covered	Energy Storage Source; Power Conversion System; DC And AC Power Distribution	bess_discharge_duration_h; bess_system_energy_density_wh_per_kg; bess_nameplate_annual_throughput_mwh	assumption_resolution: component: energy_conversion_version_transduction_percentage_inverter_power_cs_inverter; assumption_resolution: component: energy_storage_source_cell_string_lfp_prismatic_cells; assumption_resolution: component: energy_storage_source_dc_string_output_dc_contactor_or_breaker; assumption_resolution: sanity:bess_charge_rate; assumption_resolution: sanity:bess_container_energy_density; plus 24 more	accepted; accepted; accepted; accepted; accepted	none	Requirement has accepted engineering evidence in the current ledger.
Mass constraint	accepted	covered	Container Structure And Fit-Out; Maintenance And Serviceability	bess_system_energy_density_wh_per_kg	assumption_resolution: sanity:bess_container_energy_density; calculation_envelope:bess_system_energy_density_wh_per_kg; module_allocation:maintenance_serviceability; module_allocation:structure_containment; submodule_allocation:maintenance_serviceability:access_doors; plus 7 more	accepted; accepted; accepted	none	Requirement has accepted engineering evidence in the current ledger.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Power rating	accepted	covered	Power Conversion System; DC And AC Power Distribution; Control Compute And Communication	bess_discharge_duration_h	assumption_resolution: component: control_compute_communication_bms_master_controller; assumption_resolution: component: energy_conversion_transduction_plus_inverter_plus_cs_inverter; assumption_resolution: sanity:bess_c_rate; assumption_resolution: sanity:bess_container_energy_density; assumption_resolution: sanity:bess_duration; plus 20 more	accepted; accepted; accepted; accepted; accepted	none	Requirement has accepted engineering evidence in the current ledger.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 3; numeric_fact_count : 3; requirement_trace_covered: 3; requirement_trace_partial: 0; requirement_trace_uncovered: 0	capacity_mwh: 3.5 MWh; power_mw: 1 MW; mass_kg: 28000 kg; capacity_mwh: covered; power_mw: covered; mass_kg: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected energy_storage with high confidence from brief keywords.	selected_class: energy_storage; confidence: high; keyword_scores: energy_storage:6	energy_storage: 6	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 11-module energy_storage architecture from the scratch universal architecture grammar.	module_count: 11; architecture_source : scratch_universal_architecture	Energy Storage Source; Power Conversion System; Container Structure And Fit-Out; Sensing And Instrumentation; Control Compute And Communication; Safety Protection And Interlocks; plus 5 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 46; average_submodules_per_module: 4.2	energy_storage_source.cell_string; energy_storage_source.rack_structure; energy_storage_source.bms_slave; energy_storage_source.dc_string_output; energy_storage_source.module_interconnect_harness; energy_storage_source.rack_thermal_interface; plus 40 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 3; present_links: 3; missing_links: 0; graph_nodes: 57; graph_edges: 96; shared_interface_edges: 47; missing_required_graph_edges: 0	energy_storage_source -> energy_conversion_transduction via dc_bus: present; energy_storage_source -> environmental_interface via thermal_loop: present; energy_storage_source ->	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 204; critical_candidate_lines: 6; admitted_price_lines: 204	control_compute_communication via bms_network: present	Showing first 30 of 204 component candidates in the stage trace.
				lfp_prismatic_cells: LFP prismatic cells; cell_to_cell_busbar: cell-to-cell busbar; cell_terminal_hardware_set: cell terminal hardware set; cell_insulation_pad: cell insulation pad; module_fuse: module fuse; module_steel_frame: module steel frame; plus 24 more	
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 3; sanity_warn: 0; sanity_fail: 0	pass BESS C-Rate: 0.29 C; pass Rated Duration: 3.5 hours; pass Container-Level Energy Density: 125 Wh/kg	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	passed	Admitted 204 source-backed BoM line(s); remaining unpriced lines are still blocked from cost claims.	admission_status: complete; candidate_lines: 204; admitted_lines: 204; unpriced_lines: 0; unpriced_critical_lines: 0; rejected_source_records: 0	lfp_prismatic_cells: Protocol Evidence Supplier PROTOCOL-1-lfp_prismatic_cells; cell_to_cell_busbar: Protocol Evidence Supplier PROTOCOL-2-cell_to_cell_busbar; cell_terminal_hardware_set: Protocol Evidence Supplier PROTOCOL-3-cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad: Protocol Evidence Supplier PROTOCOL-4-cell_insulation_pad; module_fuse: Protocol Evidence Supplier PROTOCOL-5-module_fuse; module_steel_frame: Protocol Evidence Supplier PROTOCOL-6-module_steel_frame; plus 198 more	Cost confidence still depends on quote freshness, lead time evidence and supplier authorisation checks.

Engineering Architecture Review

Ready for BoM review

11 modules, 46 sub-modules, 204 allocated component words.

From	To	Interface	Status	Engineering Reason
energy_storage_source	energy_conversion_transduction	dc_bus	OK	Battery racks must feed the PCS over a declared DC bus.
energy_storage_source	environmental_interface	thermal_loop	OK	Cell heat must be coupled to the thermal management loop.
energy_storage_source	control_compute_communication	bms_network	OK	Cell/rack state must be visible to BMS and EMS supervision.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
11	46	47	3	0

Interface Contract Matrix

3/3 required contracts present. 16 shared interface families, 0 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
dc_bus	Energy Storage Source	Cell string; DC string output	Power Conversion System	PCS inverter bridge	present	Battery racks must feed the PCS over a declared DC bus.	Both endpoints expose the required interface through at least one submodule carrier.
thermal_loop	Energy Storage Source	Cell string; Rack thermal interface	Environmental Interface	HVAC unit; Ventilation path; Heat rejection coil	present	Cell heat must be coupled to the thermal management loop.	Both endpoints expose the required interface through at least one submodule carrier.
bms_network	Energy Storage Source	BMS slave board	Control Compute And Communication	BMS master controller	present	Cell/rack state must be visible to BMS and EMS supervision.	Both endpoints expose the required interface through at least one submodule carrier.

No local-only interface declarations in the current cross-module index.

Interface Verification Gate

accepted_interfaces

3 required interface row(s): 3 accepted, 0 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 1.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
dc_bus	Energy Storage Source	Power Conversion System	both_endpoints_carried	interface_review:energy_storage_source:energy_conversion_transduction:dc_bus	accepted	accepted	Interface contract has accepted reviewer evidence; keep evidence attached.
thermal_loop	Energy Storage Source	Environmental Interface	both_endpoints_carried	interface_review:energy_storage_source:environmental_interface:thermal_loop	accepted	accepted	Interface contract has accepted reviewer evidence; keep evidence attached.
bms_network	Energy Storage Source	Control Compute And Communication	both_endpoints_carried	interface_review:energy_storage_source:control_compute_communication:bms_network	accepted	accepted	Interface contract has accepted reviewer evidence; keep evidence attached.

Component Allocation Gate

allocation_ready

46/46 submodules ready; 0 review, 0 blocked. Component words 204; critical parts 6/6; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Energy Storage Source	Cell string	ready	5	dc_bus; thermal_loop; bms_tap_harness	lfp_prismatic_cells	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	Rack structure	ready	5	mechanical_mounts; service_access	none	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	BMS slave board	ready	5	bms_network; sensor_harness	none	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	DC string output	ready	5	dc_bus; hardwired_trip	dc_contactor_or_breaker	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	Module interconnect harness	ready	4	bms_tap_harness; sensor_harness; service_access	none	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	Rack thermal interface	ready	5	thermal_loop; mechanical_mounts; service_access	none	none	Submodule has component candidates and no allocation blocker.
Power Conversion System	PCS inverter bridge	ready	6	dc_bus; ac_bus; modbus_tcp	pcs_inverter	none	Submodule has component candidates and no allocation blocker.
Power Conversion System	Transformer stage	ready	5	ac_bus; thermal_path	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Power Conversion System	Grid filter	ready	4	ac_bus; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Power Conversion System	Grid protection relay	ready	5	ac_bus; modbus_tcp; hardwired_trip; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	ISO container shell	ready	4	lifting_points; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	Internal partitions	ready	4	fire_boundary ; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	Rack mounting rails	ready	4	mechanical_mounts	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	Cable penetration sealing	ready	5	fire_boundary ; service_accesses; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Sensing And Instrumentation	Electrical measurement	ready	4	sensor_bus; dc_bus	none	none	Submodule has component candidates and no allocation blocker.
Sensing And Instrumentation	Environmental sensors	ready	4	sensor_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Sensing And Instrumentation	Thermal sensors	ready	4	sensor_bus; thermal_loop	none	none	Submodule has component candidates and no allocation blocker.
Sensing And Instrumentation	Event recording	ready	5	sensor_bus; alarm_bus; site_scada	none	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	BMS master controller	ready	4	bms_network; hardwired_trip	bms_master_controller	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	EMS controller	ready	4	modbus_tcp; site_scada	none	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	Communications gateway	ready	4	site_scada; modbus_tcp	none	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	Cybersecurity monitoring	ready	5	site_scada; modbus_tcp; service_access	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection And Interlocks	Fire suppression system	ready	5	alarm_bus; fire_boundary	fire_detection_suppression	none	Submodule has component candidates and no allocation blocker.
Safety Protection And Interlocks	Emergency stop chain	ready	4	hardwired_trip; dc_bus; ac_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Safety Protection And Interlocks	Surge and fault protection	ready	4	dc_bus; ac_bus; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection And Interlocks	Deflagration venting	ready	5	fire_boundary ; alarm_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	HVAC unit	ready	4	electrical_power; thermal_loop; thermal_path; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	Ventilation path	ready	4	thermal_loop; thermal_path; service_access	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	Thermal insulation	ready	4	fire_boundary ; thermal_path	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	Heat rejection coil	ready	5	thermal_loop; thermal_path; electrical_power; service_access	none	none	Submodule has component candidates and no allocation blocker.
DC And AC Power Distribution	Main DC busbars	ready	4	dc_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
DC And AC Power Distribution	AC switchgear	ready	4	ac_bus; protective_earth; hardwired_trip	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
DC And AC Power Distribution	Auxiliary power distribution	ready	4	electrical_power; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
DC And AC Power Distribution	Earthing and lightning protection	ready	5	protective_earth; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Access doors and hardware	ready	4	service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Service lighting	ready	4	electrical_power; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Lifting and handling points	ready	4	lifting_points; mechanical_mounts	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Commissioning ports	ready	5	service_accesses; sensor_bus; dc_bus	none	none	Submodule has component candidates and no allocation blocker.
Coolant Circulation Process	Coolant pump station	ready	5	thermal_loop; electrical_power; sensor_bus	thermal_management_loop	none	Submodule has component candidates and no allocation blocker.
Coolant Circulation Process	Filtration and treatment	ready	4	thermal_loop; service_accesses	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Coolant Circulation Process	Expansion and air removal	ready	4	thermal_loop; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Coolant Circulation Process	Heat exchanger manifold	ready	5	thermal_loop; service_access; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Local HMI panel	ready	4	site_scada; service_access	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	External labelling	ready	4	service_access	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Operator workflow aids	ready	4	service_access; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Remote notification interface	ready	5	site_scada; alarm_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.

Module Engineering Gate

module_engineering_review_required

1/11 modules pass; 10 review, 0 blocked. Submodules/components: 46/204. Interface carriers 6/6; review questions accepted 94/94; critical unpriced lines 0 across 0 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Energy Storage Source	review	6	29	1	3/3	18/18	0	1	8 assumption row(s) need review or source evidence.
Power Conversion System	review	4	20	2	1/1	13/13	0	1	5 assumption row(s) need review or source evidence.
Container Structure And Fit-Out	review	4	17	1	0/0	7/7	0	1	1 assumption row(s) need review or source evidence.
Sensing And Instrumentation	review	4	17	0	0/0	5/5	0	0.86	1 assumption row(s) need review or source evidence.
Control Compute And Communication	review	4	17	1	1/1	10/10	0	1	4 assumption row(s) need review or source evidence.
Safety Protection And Interlocks	review	4	18	0	0/0	6/6	0	0.86	2 assumption row(s) need review or source evidence.
Environmental Interface	review	4	17	0	1/1	6/6	0	0.86	1 assumption row(s) need review or source evidence.
DC And AC Power Distribution	review	4	17	2	0/0	11/11	0	1	4 assumption row(s) need review or source evidence.

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Maintenan ce And Serviceabi lity	review	4	17	1	0/0	7/7	0	1	2 assumptio n row(s) need review or source evidence.
Coolant Circulation Process	review	4	18	0	0/0	6/6	0	0.86	2 assumptio n row(s) need review or source evidence.
Operator HMI And Labelling	pass	4	17	0	0/0	5/5	0	0.86	Module has no open determinis tic engineerin g blocker.

Module Review Roll-Up

11 ready, 0 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Energy Storage Source	ready	6	29	3/3	capacity_mwh	none	No open module-level blockers detected by current deterministic checks.
Power Conversion System	ready	4	20	1/1	capacity_mwh ; power_mw	none	No open module-level blockers detected by current deterministic checks.
Container Structure And Fit-Out	ready	4	17	0/0	mass_kg	none	No open module-level blockers detected by current deterministic checks.
Sensing And Instrumentation	ready	4	17	0/0	none	none	No open module-level blockers detected by current deterministic checks.
Control Compute And Communication	ready	4	17	1/1	power_mw	none	No open module-level blockers detected by current deterministic checks.
Safety Protection And Interlocks	ready	4	18	0/0	none	none	No open module-level blockers detected by current deterministic checks.
Environmental Interface	ready	4	17	1/1	none	none	No open module-level blockers detected by current deterministic checks.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
DC And AC Power Distribution	ready	4	17	0/0	capacity_mwh ; power_mw	none	No open module-level blockers detected by current deterministic checks.
Maintenance And Serviceability	ready	4	17	0/0	mass_kg	none	No open module-level blockers detected by current deterministic checks.
Coolant Circulation Process	ready	4	18	0/0	none	none	No open module-level blockers detected by current deterministic checks.
Operator HMI And Labelling	ready	4	17	0/0	none	none	No open module-level blockers detected by current deterministic checks.

Engineering Review Pack

81 reviewer questions: 78 accepted, 0 ready, 3 needing review, 0 blocked. Coverage: 11 modules, 46 submodules, 3 interfaces, 3 calculations, 18 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Annual MWh delivered	Can the Annual MWh delivered assumption be accepted for architecture review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performance claim.	none	none	none	none
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	accepted	G99	Can the G99 assumption be accepted for architecture review, or must it be replaced with evidence?	Protection settings, relay test certificates and commissioning records.	none	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumption_resolution	major	accepted	IEC-62619	Can the IEC-62619 assumption be accepted for architecture review, or must it be replaced with evidence?	Cell certificates and pack-level safety validation.	none	none	none	none
assumption_resolution	major	accepted	UL-9540A	Can the UL-9540A assumption be accepted for architecture review, or must it be replaced with evidence?	Cell, module and unit-level propagation evidence.	none	none	none	none
assumption_resolution	major	accepted	BMS master controller	Can the BMS master controller assumption be accepted for architecture review, or must it be replaced with evidence?	Review admitted source evidence and component identity before procurement.	control_compute_communication	none	none	none
assumption_resolution	major	accepted	PCS inverter	Can the PCS inverter assumption be accepted for architecture review, or must it be replaced with evidence?	Review admitted source evidence and component identity before procurement.	energy_conversion_transduction	none	none	none
assumption_resolution	major	accepted	LFP prismatic cells	Can the LFP prismatic cells assumption	Review admitted source evidence and	energy_storage_source	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				n be accepted for architecture review, or must it be replaced with evidence?	component identity before procurement.				
assumption_resolution	major	accepted	DC contactor or breaker	Can the DC contactor or breaker assumption be accepted for architecture review, or must it be replaced with evidence?	Review admitted source evidence and component identity before procurement.	energy_storage_source	none	none	none
assumption_resolution	major	accepted	Thermal management loop	Can the Thermal management loop assumption be accepted for architecture review, or must it be replaced with evidence?	Review admitted source evidence and component identity before procurement.	mass_fluid_transport_process	none	none	none
assumption_resolution	major	accepted	Fire detection/suppression	Can the Fire detection/suppression assumption be accepted for architecture review, or must it be replaced with evidence?	Review admitted source evidence and component identity before procurement.	safety_protection	none	none	none
assumption_resolution	major	accepted	energy_storage_source -> control_compute_co	Can the energy_storage_source -> control_co	Review the interface carrier, direction,	none	bms_network	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
			communication	compute_communication assumption be accepted for architecture review, or must it be replaced with evidence?	capacity, safety margin and failure mode before detailed design or sourcing.				
assumption_resolution	major	accepted	energy_storage_source -> energy_conversion_transduction	Can the energy_storage_source -> energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	dc_bus	none	none
assumption_resolution	major	accepted	energy_storage_source -> environmental_interface	Can the energy_storage_source -> environmental_interface assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	thermal_loop	none	none
assumption_resolution	major	accepted	BESS C-Rate	Can the BESS C-Rate assumption be accepted for architecture review, or must it be	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale	energy_storage_source; energy_conversion_transduction; power_distribution; control_compute_co	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				replaced with evidence?	for this envelope.	mmunicati on			
assumptio n_resoluti on	major	accepted	Container-Level Energy Density	Can the Container-Level Energy Density assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_storage_source; energy_conversion_transduction; power_distribution; structure_containment; maintenance_serviceability	none	none	none
assumptio n_resoluti on	major	accepted	Rated Duration	Can the Rated Duration assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation , reviewer note, test plan or datasheet -backed rationale for this envelope.	energy_storage_source; energy_conversion_transduction; power_distribution; control_compute_communication	none	none	none
calculation_envelope	major	accepted	BESS discharge duration	Does the BESS discharge duration calculation support the architecture envelope, or does the concept need revision?	Confirm use case, PCS rating, usable energy window, C-rate limits and warranty throughput.	energy_storage_source; energy_conversion_transduction; power_distribution; control_compute_communication	none	bess_discharge_duration_h	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
calculation_envelope	major	accepted	BESS annual nameplate throughput	Does the BESS annual nameplate throughput calculation support the architecture envelope, or does the concept need revision?	Confirm cycles per year, usable depth of discharge, round-trip efficiency and degradation assumptions.	energy_storage_source; energy_conversion_transduction; power_distribution	none	bess_nameplate_annual_throughput_mwh	none
calculation_envelope	major	accepted	BESS system energy density	Does the BESS system energy density calculation support the architecture envelope, or does the concept need revision?	Review cell mass, racks, enclosure, HVAC, fire protection , transformer/PCS boundary and shipping mass allowance.	energy_storage_source; energy_conversion_transduction; power_distribution; structure_containment; maintenance_serviceability	none	bess_system_energy_density_Wh_per_kg	none
interface_contract	major	accepted	Energy Storage Source -> Control Compute And Communication via bms_network	Is the bms_network interface between Energy Storage Source and Control Compute And Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; control_compute_communication	bms_network	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	accepted	Energy Storage Source -> Power Conversion System via dc_bus	Is the dc_bus interface between Energy Storage Source and Power Conversio	Interface-review evidence naming both endpoint owners, carrier submodule	energy_storage_source; energy_conversion_transduction	dc_bus	none	Both endpoints expose the required interface through at least one

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				n System correctly owned and physically carried?	es, capacity limits and any deferred checks.				submodule carrier.
interface_contract	major	accepted	Energy Storage Source -> Environmental Interface via thermal_loop	Is the thermal_loop interface between Energy Storage Source and Environmental Interface correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; environmental_interface	thermal_loop	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	accepted	Control Compute And Communication	Does Control Compute And Communication have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	bms_network; modbus_tcp; sensor_buses; alarm_bus; site_scada; service_access; hardwired_trip	bess_capacity; bess_duration	none
module_allocation	major	accepted	Power Conversion System	Does Power Conversion System have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	energy_conversion_transduction	dc_bus; ac_bus; modbus_tcp; thermal_path; hardwired_trip; protective_earth	bess_capacity; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	accepted	Energy Storage Source	Does Energy Storage Source have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	energy_storage_source	dc_bus; thermal_load; bms_network; mechanical_mounts; bms_tap_harness; service_access; sensor_harness; hardwired_trip	bess_container; bess_duration; bess_container_energy_density	none
module_all ocation	major	accepted	Environmental Interface	Does Environmental Interface have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	environmental_interface	thermal_load; thermal_path; electrical_power; sensor_bus; service_access; fire_boundary	none	none
module_all ocation	major	accepted	Operator HMI And Labelling	Does Operator HMI And Labelling have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	hmi_ergonomics	site_scada; service_access; alarm_bus	none	none
module_all ocation	major	accepted	Maintenance And Serviceability	Does Maintenance And Serviceability have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	maintenance_serviceability	service_access; lifting_points; protective_earth; sensor_bus; dc_bus; electrical_power; mechanical_mounts	bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	accepted	Coolant Circulation Process	Does Coolant Circulation Process have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	mass_fluid _transport _process	thermal_lo op; sensor_bu s; service_ac cess; electrical_ power	none	none
module_all ocation	major	accepted	DC And AC Power Distribution	Does DC And AC Power Distribution have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	power_dis tribution	dc_bus; ac_bus; electrical_ power; protective _earth; hardwired _trip; service_ac cess	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none
module_all ocation	major	accepted	Safety Protection And Interlocks	Does Safety Protection And Interlocks have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	safety_pro tection	alarm_bus ; hardwired _trip; fire_bound ary; dc_bus; ac_bus; service_ac cess; protective _earth	none	none
module_all ocation	major	accepted	Sensing And Instrument ation	Does Sensing And Instrument ation have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	sensing_in strumenta tion	sensor_bu s; alarm_bus ; bms_netw ork; site_scada ; dc_bus; thermal_lo op	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
module_all ocation	major	accepted	Container Structure And Fit-Out	Does Container Structure And Fit-Out have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	structure_containme nt	mechanical_mounts; service_access; fire_boundary; lifting_points; protective_earth	bess_container_energy_density	none
submodule_allocation	major	accepted	Control Compute And Communication / BMS master controller	Do the component candidate s and interfaces assigned to BMS master controller make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communicati on	bms_network; hardwired_trip	bess_c_rate; bess_duration	none
submodule_allocation	major	accepted	Control Compute And Communication / Communications gateway	Do the component candidate s and interfaces assigned to Communications gateway make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communicati on	site_scada ; modbus_tcp	bess_c_rate; bess_duration	none
submodule_allocation	major	accepted	Control Compute And Communication / Cybersecurity monitoring	Do the component candidate s and interfaces assigned to Cybersecurity monitoring make engineering	Reviewer note confirming the submodule role, component allocation and any missing carriers or	control_compute_communicati on	site_scada ; modbus_tcp; service_access	bess_c_rate; bess_duration	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				g sense for its stated purpose?	capacity checks.				
submodule_allocation	major	accepted	Control Compute And Communication / EMS controller	Do the component candidate s and interfaces assigned to EMS controller make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	control_compute_communication	modbus_tcp; site_scada	bess_capacity; bess_duration	none
submodule_allocation	major	accepted	Power Conversion System / Grid filter	Do the component candidate s and interfaces assigned to Grid filter make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	ac_bus; protective_earth	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Power Conversion System / Grid protection relay	Do the component candidate s and interfaces assigned to Grid protection relay make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	ac_bus; modbus_tcp; hardwired_trip; protective_earth	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Power Conversion System / PCS inverter bridge	Do the component candidate s and interfaces assigned to PCS inverter bridge make engineering sense	Reviewer note confirming the submodule role, component allocation and any missing carriers or	energy_conversion_transduction	dc_bus; ac_bus; modbus_tcp	bess_capacity; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for its stated purpose?	capacity checks.				
submodule_allocation	major	accepted	Power Conversion System / Transformer stage	Do the component candidates and interfaces assigned to Transformer stage make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	ac_bus; thermal_path	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Energy Storage Source / BMS slave board	Do the component candidates and interfaces assigned to BMS slave board make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	bms_network; sensor_harness	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Energy Storage Source / Cell string	Do the component candidates and interfaces assigned to Cell string make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	dc_bus; thermal_loop; bms_tap_harness	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Energy Storage Source / DC string output	Do the component candidates and interfaces assigned to DC string output make	Reviewer note confirming the submodule role, component allocation and any missing	energy_storage_source	dc_bus; hardwired_trip	bess_capacity; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	accepted	Energy Storage Source / Module interconnect harness	engineering sense for its stated purpose?	carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Module interconnect harness make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	bms_tap_harness; sensor_harness; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Energy Storage Source / Rack structure	Do the component candidate s and interfaces assigned to Rack structure make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	mechanical_mounts; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Energy Storage Source / Rack thermal interface	Do the component candidate s and interfaces assigned to Rack thermal interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	thermal_loop; mechanical_mounts; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	Environmental Interface / Heat rejection coil	Do the component candidate s and interfaces assigned to Heat rejection	Reviewer note confirming the submodule role, component allocation	environmental_interface	thermal_loop; thermal_path; electrical_power; service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				coil make engineering sense for its stated purpose?	and any missing carriers or capacity checks.				
submodule_allocation	major	accepted	Environmental Interface / HVAC unit	Do the component candidates and interfaces assigned to HVAC unit make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	electrical_power; thermal_loop; thermal_path; sensor_buses	none	none
submodule_allocation	major	accepted	Environmental Interface / Thermal insulation	Do the component candidates and interfaces assigned to Thermal insulation make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	fire_boundary; thermal_path	none	none
submodule_allocation	major	accepted	Environmental Interface / Ventilation path	Do the component candidates and interfaces assigned to Ventilation path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	thermal_loop; thermal_path; service_access	none	none
submodule_allocation	major	accepted	Operator HMI And Labelling / External labelling	Do the component candidates and interfaces assigned to External labelling make engineering	Reviewer note confirming the submodule role, component allocation and any missing	hmi_ergonomics	service_access	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	accepted	Operator HMI And Labelling / Local HMI panel	g sense for its stated purpose?	carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Local HMI panel make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	site_scada ; service_access	none	none
				Do the component candidate s and interfaces assigned to Operator workflow aids make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	service_access; alarm_bus	none	none
				Do the component candidate s and interfaces assigned to Remote notification interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	site_scada ; alarm_bus ; service_access	none	none
				Do the component candidate s and interfaces assigned to Access doors and hardware make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any	maintenance_serviceability	service_access	bess_controller_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				make engineering sense for its stated purpose?	missing carriers or capacity checks.				
submodule_allocation	major	accepted	Maintenance And Serviceability / Commissioning ports	Do the component candidate s and interfaces assigned to Commissioning ports make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	service_access; sensor_buss; dc_bus	bess_container_energy_density	none
submodule_allocation	major	accepted	Maintenance And Serviceability / Lifting and handling points	Do the component candidate s and interfaces assigned to Lifting and handling points make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	lifting_points; mechanical_mounts	bess_container_energy_density	none
submodule_allocation	major	accepted	Maintenance And Serviceability / Service lighting	Do the component candidate s and interfaces assigned to Service lighting make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_service_ability	electrical_power; service_access	bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	accepted	Coolant Circulation Process / Expansion and air removal	Do the component candidates and interfaces assigned to Expansion and air removal make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	thermal_loop; sensor_buses	none	none
submodule_allocation	major	accepted	Coolant Circulation Process / Filtration and treatment	Do the component candidates and interfaces assigned to Filtration and treatment make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	thermal_loop; service_access	none	none
submodule_allocation	major	accepted	Coolant Circulation Process / Heat exchanger manifold	Do the component candidates and interfaces assigned to Heat exchanger manifold make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	thermal_loop; service_access; sensor_buses	none	none
submodule_allocation	major	accepted	Coolant Circulation Process / Coolant pump station	Do the component candidates and interfaces assigned to Coolant pump station make	Reviewer note confirming the submodule role, component allocation and any missing	mass_fluid_transport_process	thermal_loop; electrical_power; sensor_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				engineering sense for its stated purpose?	carriers or capacity checks.				
submodule_allocation	major	accepted	DC And AC Power Distribution / AC switchgear	Do the component candidates and interfaces assigned to AC switchgear make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	ac_bus; protective_earth; hardwired_trip	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	DC And AC Power Distribution / Auxiliary power distribution	Do the component candidates and interfaces assigned to Auxiliary power distribution make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	electrical_power; protective_earth	bess_capacity; bess_duration; bess_container_energy_density	none
submodule_allocation	major	accepted	DC And AC Power Distribution / Main DC busbars	Do the component candidates and interfaces assigned to Main DC busbars make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	dc_bus; service_access	bess_capacity; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodul e_allocation	major	accepted	DC And AC Power Distribution / Earthing and lightning protection	Do the componen t candidate s and interfaces assigned to Earthing and lightning protection make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	power_dis tribution	protective _earth; service_ac cess	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none
submodul e_allocation	major	accepted	Safety Protection And Interlocks / Deflagrati on venting	Do the componen t candidate s and interfaces assigned to Deflagrati on venting make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	safety_pro tection	fire_bound ary; alarm_bus ; service_ac cess	none	none

Showing first 70 of 81 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
BESS C-Rate	pass	0.29 C	0.1C to 1.0C for typical grid-storage duration systems	Power-to-energy ratio implies 0.29C, equivalent to roughly 3.5 hours at rated power.
Rated Duration	pass	3.5 hours	1 to 6 hours for common containerised BESS applications	3.5 hours is plausible for a containerised grid-support BESS before detailed cell count and PCS sizing.
Container-Level Energy Density	pass	125 Wh/kg	50 to 200 Wh/kg at container/system level before detailed structural mass review	125 Wh/kg is within a plausible container-level range, but it still needs a mass roll-up by rack, PCS, HVAC and enclosure.

Engineering Calculation Ledger

3 calculations: 2 within deterministic envelope, 1 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
BESS discharge duration	within envelope	capacity_mwh / power_mw	3.5 h	2 to 8 h is a common grid-storage concept envelope before duty-cycle proof.	3.5 h implied by nameplate capacity and PCS power.	Confirm use case, PCS rating, usable energy window, C-rate limits and warranty throughput.
BESS system energy density	within envelope	(capacity_mwh * 1,000,000) / mass_kg	125 Wh/kg	60 to 180 Wh/kg is a broad containerised LFP system envelope before packaging proof.	125 Wh/kg implied by energy and gross mass.	Review cell mass, racks, enclosure, HVAC, fire protection, transformer/PCS boundary and shipping mass allowance.
BESS annual nameplate throughput	needs review	capacity_mwh * 365	1277.5 MWh/year	Arithmetic only; real throughput depends on dispatch, degradation, auxiliary load and warranty.	1277.5 MWh/year assumes one full equivalent cycle per day.	Confirm cycles per year, usable depth of discharge, round-trip efficiency and degradation assumptions.

Engineering Assumption Ledger

21 assumptions: 3 brief-supported, 9 model-present, 9 needing engineering review, 0 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 0.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Energy capacity	brief_requirement	brief supported	no	no	Energy capacity = 3.5 MWh is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Power rating	brief_requirement	brief supported	no	no	Power rating = 1 MW is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Mass constraint	brief_requirement	brief supported	no	no	Mass constraint = 28000 kg is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Annual MWh delivered	derived_metric	review required	no	no	Annual MWh delivered uses capacity_mwh * cycles_per_year * efficiency with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	review required	no	no	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	review required	no	no	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
BESS C-Rate	sanity_envelope	review required	no	no	BESS C-Rate currently evaluates as pass against 0.1C to 1.0C for typical grid-storage duration systems.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Rated Duration	sanity_envelope	review required	no	no	Rated Duration currently evaluates as	Supply an engineering calculation,

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					pass against 1 to 6 hours for common containerised BESS applications.	reviewer note, test plan or datasheet-backed rationale for this envelope.
Container-Level Energy Density	sanity_envelope	review required	no	no	Container-Level Energy Density currently evaluates as pass against 50 to 200 Wh/kg at container/system level before detailed structural mass review.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
energy_storage_source -> energy_conversion_transduction	interface_closure	model present	no	no	dc_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
energy_storage_source -> environmental_interface	interface_closure	model present	no	no	thermal_loop is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
energy_storage_source -> control_compute_communication	interface_closure	model present	no	no	bms_network is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
LFP prismatic cells	critical_component	model present	no	no	LFP prismatic cells is a critical architecture-derived component candidate with quantity 1 candidate.	Review admitted source evidence and component identity before procurement.
DC contactor or breaker	critical_component	model present	no	no	DC contactor or breaker is a critical architecture-derived component	Review admitted source evidence and component identity before procurement.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					candidate with quantity 1 candidate.	
PCS inverter	critical_component	model present	no	no	PCS inverter is a critical architecture-derived component candidate with quantity 1 candidate.	Review admitted source evidence and component identity before procurement.
BMS master controller	critical_component	model present	no	no	BMS master controller is a critical architecture-derived component candidate with quantity 1 candidate.	Review admitted source evidence and component identity before procurement.
Fire detection/suppression	critical_component	model present	no	no	Fire detection/suppression is a critical architecture-derived component candidate with quantity 1 candidate.	Review admitted source evidence and component identity before procurement.
Thermal management loop	critical_component	model present	no	no	Thermal management loop is a critical architecture-derived component candidate with quantity 1 candidate.	Review admitted source evidence and component identity before procurement.
IEC-62619	compliance_evidence	review required	no	no	Industrial lithium battery safety is relevant to this product class in Global / UK-adopted.	Cell certificates and pack-level safety validation.
UL-9540A	compliance_evidence	review required	no	no	Thermal runaway fire propagation test is relevant to this product class in US / investor diligence.	Cell, module and unit-level propagation evidence.
G99	compliance_evidence	review required	no	no	UK grid connection requirements is relevant to this	Protection settings, relay test certificates and

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					product class in UK.	commissioning records.

Engineering Verification Plan

25 activities: 20 ready for review, 5 open, 0 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Energy Storage Source architecture allocation	ready	design_review	Energy Storage Source	capacity_mwh	dc_bus; thermal_loop; bms_network; mechanical_mounts; bms_tap_harness; service_access; sensor_harness; hardwired_trip	lfp_prismatic_cells; cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad; module_fuse; module_steel_frame; compression_plate; compression_tie_rod_set; plus 21 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Energy Storage Source requirement calculations	ready	calculation	Energy Storage Source	capacity_mwh	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Power Conversion System architecture allocation	ready	design_review	Power Conversion System	capacity_mwh; power_mw	dc_bus; ac_bus; modbus_tcp; thermal_path; hardwired_trip; protective_earth	pcs_inverter; igbt_power_module; dc_link_capacitor_bank; gate_driver_board; inverter_control_board; liquid_cooled_heatsink; cast_resin_transformer; hv_winding;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						plus 12 more	where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify Power Conversion System requirement calculations	ready	calculation	Power Conversion System	capacity_mwh; power_mw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Container Structure And Fit-Out architecture allocation	ready	design_review	Container Structure And Fit-Out	mass_kg	mechanical_mounts; service_access; fire_boundary; lifting_points; protective_earth	corten_steel_side_panel; corner_casting; marine_plywood_floor; roof_panel; steel_stud_frame; fire_rated_boards; mineral_wool_insulation; intumescent_sealant; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Container Structure And Fit-Out	ready	calculation	Container Structure And Fit-Out	mass_kg	none	none	Each linked requirement has a calculation,	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
requirement calculations							sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Review Sensing And Instrumentation architecture allocation	ready	design_review	Sensing And Instrumentation	none	sensor_bus; alarm_bus; bms_network; site_scada; dc_bus; thermal_loop	dc_voltage_transducer; hall_current_sensor; insulation_monitoring_device; auxiliary_metering_transducer; temperature_humidity_sensor; hydrogen_detector; smoke_aspirating_detector; door_position_switch; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Control Compute And Communication architecture allocation	ready	design_review	Control Compute And Communication	power_mw	bms_network; modbus_tcp; sensor_bus; alarm_bus; site_scada; service_access; hardwired_trip	bms_master_controller; isolation_interface; contactor_driver_output; ; soc_estimation_firmware; industrial_controller; i_o_expansion_module; network_switch; cellular_router; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							No blocker or major architecture issue is attached to the module.	
Verify Control Compute And Communication requirement calculations	ready	calculation	Control Compute And Communication	power_mw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Safety Protection And Interlocks architecture allocation	ready	design_review	Safety Protection And Interlocks	none	alarm_bus; hardwired_trip; fire_boundaries; dc_bus; ac_bus; service_access; protective_earth	fire_detection_suppression; clean_agent_cylinder; release_solenoid; nozzle_pipe work; pressure_switch; e_stop_mushroom_button; safety_relay; dc_shunt_trip_output; plus 10 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Environmental Interface architecture allocation	ready	design_review	Environmental Interface	none	thermal_loop; thermal_patching; electrical_power; sensor_bus; service_access;	container_hvac_unit; air_filter_element; condensate_drain_kit; hvac_controller; intake_louver	Every submodule has a stated purpose and at least one component candidate. Interfaces	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
					fire_boundary	e; exhaust_louvre; ec_fan_assembly; airflow_baffle; plus 9 more	named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Review DC And AC Power Distribution architecture allocation	ready	design_review	DC And AC Power Distribution	capacity_mwh; power_mw	dc_bus; ac_bus; electrical_power; protective_earth; hardwired_trip; service_access	copper_dc_busbar; busbar_support_insulator; busbar_shroud; torque_witness_marker; moulded_case_circuit_breaker; ac_contactor; metering_controller_set; terminal_block_assembly; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify DC And AC Power Distribution requirement calculations	ready	calculation	DC And AC Power Distribution	capacity_mwh; power_mw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							or test evidence.	
Review Maintenance And Serviceability architecture allocation	ready	design_review	Maintenance And Serviceability	mass_kg	service_access; lifting_points; protective_earth; sensor_bus; dc_bus; electrical_power; mechanical_mounts	lockable_access_door; door_seal_gasket; panic_release_latch; door_stay_arm; led_service_luminaire; emergency_light_fitting; maintenance_receptacle; lighting_switchgear; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Maintenance And Serviceability requirement calculations	ready	calculation	Maintenance And Serviceability	mass_kg	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Coolant Circulation Process architecture allocation	ready	design_review	Coolant Circulation Process	none	thermal_loop; sensor_bus; service_access; electrical_power	thermal_management_loop; coolant_circulation_pump; pump_vfd; check_valve; isolation_valve; coolant_filter_housing;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						replaceable_filter_cartridge; glycol_concentration_sensor; plus 10 more	responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Review Operator HMI And Labelling architecture allocation	ready	design_review	Operator HMI And Labelling	none	site_scada; service_access; alarm_bus	industrial_touchscreen; panel_pc; key_switch; alarm_buzzer_beacon; high_voltage_warning_label; arc_flash_boundary_label; fire_suppression_instruction_placard; asset_qr_label; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify dc_bus interface contract	ready	interface_review	Energy Storage Source -> Power Conversion System	none	dc_bus	lfp_prismatic_cells; cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad; module_fuse; dc_contactor_or_breaker; precharge_resistor; hv_fuse_holder; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Verify thermal_loop interface contract	ready	interface_review	Energy Storage Source -> Environmental Interface	none	thermal_loop	lfp_prismatic_cells; cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad; module_fuse; rack_cold_plate_manifold; thermally_conductive_gap_pad; coolant_quick_connect_or_pair; plus 15 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify bms_network interface contract	ready	interface_review	Energy Storage Source -> Control Compute And Communication	none	bms_network	bms_slave_pcb; afe_monitor_ic; ntc_thermistors_harness; isolated_cann_transceiver; conformal_coating; bms_master_controller; isolation_interface; contactor_driver_output; plus 1 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Plan evidence for IEC-62619	open	compliance_review	System Compliance	none	none	none	Cell certificates and pack-level safety validation.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for UL-9540A	open	compliance_review	System Compliance	none	none	none	Cell, module and unit-level propagation evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for G99	open	compliance_review	System Compliance	none	none	none	Protection settings, relay test certificates	Compliance evidence not yet collected in

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and commissioning records.	scratch prototype.
Review risk control for Thermal runaway propagation	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.
Review risk control for DC arc fault	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

25 accepted, 0 rejected, 0 deferred, 0 pending, 0 blocked without evidence. Review coverage 1; acceptance ratio 1. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Energy Storage Source architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:energy_storage_source	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify Energy Storage Source requirement calculations	accepted	calculation	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/calculation:energy_storage_source	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Power Conversion System architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:energy_conversion_transduction	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify Power Conversion System requirement calculations	accepted	calculation	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/calculation:energy_conversion_transduction	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Container Structure And Fit-Out architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:structure_containment	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify Container Structure And Fit-Out requirement calculations	accepted	calculation	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/calculation:structure_containment	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Sensing And Instrumentation architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:sensing_instrumentation	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Control Compute And Communication architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:control_compute_communication	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify Control Compute And Communication requirement calculations	accepted	calculation	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/calculation:control_compute_communication	Accepted by reviewer; keep the evidence reference attached to the activity.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Safety Protection And Interlocks architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:safety_protection	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Environmental Interface architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:environmental_interface	Accepted by reviewer; keep the evidence reference attached to the activity.
Review DC And AC Power Distribution architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:power_distribution	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify DC And AC Power Distribution requirement calculations	accepted	calculation	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/calculation:power_distribution	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Maintenance And Serviceability architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:maintenance_serviceability	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify Maintenance And Serviceability requirement calculations	accepted	calculation	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/calculation:maintenance_serviceability	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Coolant Circulation Process architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:mass_fluid_transport_processes	Accepted by reviewer; keep the evidence reference attached to the activity.
Review Operator HMI And Labelling architecture allocation	accepted	design_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/design_review:hmi_ergonomics	Accepted by reviewer; keep the evidence reference attached to the activity.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify dc_bus interface contract	accepted	interface_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/interface_review:energy_storage_source:energy_conversion_transduction:dc_bus	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify thermal_loop interface contract	accepted	interface_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/interface_review:energy_storage_source:environmental_interface:thermal_loop	Accepted by reviewer; keep the evidence reference attached to the activity.
Verify bms_network interface contract	accepted	interface_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/interface_review:energy_storage_source:control_compute_communication:bms_network	Accepted by reviewer; keep the evidence reference attached to the activity.
Plan evidence for IEC-62619	accepted	compliance_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/compliance_review:IEC-62619	Accepted by reviewer; keep the evidence reference attached to the activity.
Plan evidence for UL-9540A	accepted	compliance_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/compliance_review:UL-9540A	Accepted by reviewer; keep the evidence reference attached to the activity.
Plan evidence for G99	accepted	compliance_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/compliance_review:G99	Accepted by reviewer; keep the evidence reference attached to the activity.
Review risk control for Thermal runaway propagation	accepted	compliance_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/risk_review:1	Accepted by reviewer; keep the evidence reference attached to the activity.
Review risk control for DC arc fault	accepted	compliance_review	Protocol Evidence Reviewer	test-fixture://review-evidence-closure/review/risk_review:2	Accepted by reviewer; keep the evidence reference attached to the activity.

Energy Storage Source

Stores roughly 3.5 MWh usable energy in rack-mounted LFP cell assemblies.

- dc_bus
- thermal_loop
- bms_network
- mechanical_mounts

Cell string

Electrochemical storage path assembled from series/parallel LFP cells.

Components: LFP prismatic cells, cell-to-cell busbar, cell terminal hardware set, cell insulation pad, module fuse

Rack structure

Constrains cell modules during transport, vibration and cycling expansion.

Components: module steel frame, compression plate, compression tie rod set, module top cover, rack label plate

BMS slave board

Measures cell voltage and temperature at rack level.

Components: BMS slave PCB, AFE monitor IC, NTC thermistor harness, isolated CAN transceiver, conformal coating

DC string output

Combines rack outputs into a protected high-voltage DC path.

Components: DC contactor or breaker, precharge resistor, HV fuse holder, pack current sensor, service disconnect

Module interconnect harness

Routes low-voltage sensing, communications and temperature signals across rack assemblies.

Components: cell voltage sense loom, rack CAN drop cable, harness strain relief rail, low-smoke cable sleeve

Rack thermal interface

Couples cell modules to the container thermal loop while preserving service replacement access.

Components: rack cold plate manifold, thermally conductive gap pad, coolant quick connector pair, rack drip tray, leak detection rope

Power Conversion System

Converts the battery DC bus into 1 MW grid-compatible AC output.

- dc_bus
- ac_bus
- modbus_tcp
- thermal_path
- hardwired_trip
- protective_earth

PCS inverter bridge

Performs bidirectional DC/AC conversion.

Components: PCS inverter, IGBT power module, DC link capacitor bank, gate driver board, inverter control board, liquid-cooled heatsink

Transformer stage

Adapts PCS output voltage to site connection voltage.

Components: cast resin transformer, HV winding, LV winding, temperature probe, anti-vibration mount

Grid filter

Controls harmonics and conducted emissions.

Components: grid filter choke, AC filter capacitor, EMI filter assembly, protective earth bar

Grid protection relay

Supervises grid voltage, frequency and trip settings at the point of connection.

Components: grid protection relay, voltage sensing fuse set, relay test block, trip circuit monitor, G99 settings label

Container Structure And Fit-Out

Provides weatherproof containment, transport structure and internal segregation.

- mechanical_mounts
- service_access
- fire_boundary
- lifting_points
- protective_earth

ISO container shell

Forms the weather-tight transportable enclosure.

Components: corten steel side panel, corner casting, marine plywood floor, roof panel

Internal partitions

Separates battery, PCS, controls and service zones.

Components: steel stud frame, fire-rated board, mineral wool insulation, intumescent sealant

Rack mounting rails

Anchors battery racks against transport and seismic loads.

Components: floor reinforcement plate, rack mounting rail, vibration isolator, anchor bolt set

Cable penetration sealing

Seals power, control and auxiliary cable entries through container fire and weather boundaries.

Components: gland plate assembly, firestop collar set, EMC cable gland, rain drip shield, penetration schedule label

Sensing And Instrumentation

Measures electrical, thermal and environmental state for operation and safety.

- sensor_bus
- alarm_bus
- bms_network
- site_scada

Electrical measurement

Measures voltage, current and insulation state.

Components: DC voltage transducer, Hall current sensor, insulation monitoring device, auxiliary metering transducer

Environmental sensors

Measures container air conditions and gas hazards.

Components: temperature humidity sensor, hydrogen detector, smoke aspirating detector, door position switch

Thermal sensors

Tracks coolant and heat exchanger temperatures.

Components: coolant inlet thermistor, coolant outlet thermistor, flow meter, pressure transducer

Event recording

Captures timestamped trips, alarms and measurements for engineering review.

Components: fault recorder module, event timestamp logger, digital input card, local historian buffer, calibration certificate holder

Control Compute And Communication

Coordinates BMS, PCS, auxiliaries and site-level dispatch.

bms_network	modbus_tcp	sensor_bus	alarm_bus	site_scada	service_access
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BMS master controller

Supervises pack state and protective limits.

Components: BMS master controller, isolation interface, contactor driver output, SOC estimation firmware

EMS controller

Dispatches power and monitors system state.

Components: industrial controller, I/O expansion module, network switch, cellular router

Communications gateway

Bridges local controls to site and remote systems.

Components: Modbus TCP gateway, SCADA protocol adapter, time synchronisation module, cybersecurity firewall

Cybersecurity monitoring

Manages secure access, logging and recovery data for remote operation.

Components: VPN access appliance, syslog collector, account key switch, firmware escrow tag, configuration backup module

Safety Protection And Interlocks

Detects unsafe conditions and forces the system into a protected state.

alarm_bus	hardwired_trip	fire_boundary	dc_bus	ac_bus	service_access
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Fire suppression system

Detects and suppresses incipient battery-container fire.

Components: Fire detection/suppression, clean-agent cylinder, release solenoid, nozzle pipework, pressure switch

Emergency stop chain

Trips high-energy paths through hardwired safety logic.

Components: E-stop mushroom button, safety relay, DC shunt trip output, AC breaker trip coil

Surge and fault protection

Limits transient and overcurrent damage.

Components: DC surge protection device, AC surge protection device, arc flash label set, protective earth bonding kit

Deflagration venting

Provides a controlled pressure relief path if battery off-gas ignition occurs.

Components: pressure relief vent panel, vent microswitch, flame arrestor mesh, vent weather hood, post-event inspection tag

Environmental Interface

Maintains the container operating envelope across weather and heat loads.

- thermal_loop
- thermal_path
- electrical_power
- sensor_bus
- service_access

HVAC unit

Controls air temperature and humidity inside compartments.

Components: container HVAC unit, air filter element, condensate drain kit, HVAC controller

Ventilation path

Directs airflow around PCS, controls and service areas.

Components: intake louver, exhaust louver, EC fan assembly, airflow baffle

Thermal insulation

Reduces ambient heat transfer into the container.

Components: insulation panel, vapour barrier film, thermal break strip, seal gasket set

Heat rejection coil

Rejects battery and PCS heat from the coolant loop to outdoor air.

Components: dry cooler coil, EC condenser fan, fan contactor, coil guard grille, drain-safe condensate pan

DC And AC Power Distribution

Routes high-voltage DC, grid AC and auxiliary power through protected paths.

- dc_bus
- ac_bus
- electrical_power
- protective_earth
- hardwired_trip

Main DC busbars

Carries pack current from racks to PCS.

Components: copper DC busbar, busbar support insulator, busbar shroud, torque witness marker

AC switchgear

Connects PCS output to site AC connection.

Components: moulded-case circuit breaker, AC contactor, metering CT set, terminal block assembly

Auxiliary power distribution

Supplies controls, HVAC, pumps and service loads.

Components: auxiliary transformer, 24 V DC power supply, miniature circuit breaker bank, UPS module

Earthing and lightning protection

Bonds exposed metalwork and surge devices to a verifiable protective-earth network.

Components: main earth bar, earth stud array, lightning surge counter, equipotential bonding strap, earth continuity test point

Maintenance And Serviceability

Lets technicians inspect, isolate, lift and replace internal assemblies safely.

- service_access
- lifting_points
- protective_earth
- sensor_bus
- dc_bus

Access doors and hardware

Provides controlled entry to service zones.

Components: lockable access door, door seal gasket, panic release latch, door stay arm

Service lighting

Illuminates work zones during inspection.

Components: LED service luminaire, emergency light fitting, maintenance receptacle, lighting switchgear

Lifting and handling points

Supports transport, installation and module replacement.

Components: certified lifting lug, forklift pocket reinforcement, removable rack guide, service trolley interface

Commissioning ports

Provides controlled access points for electrical, coolant and controller commissioning checks.

Components: insulated test socket, coolant sampling port, laptop service port, lockout hasp rail, commissioning torque log plate

Coolant Circulation Process

Moves coolant through battery racks, heat exchangers and serviceable filters.

- thermal_loop
- sensor_bus
- service_access
- electrical_power

Coolant pump station

Circulates coolant through rack cold plates.

Components: Thermal management loop, coolant circulation pump, pump VFD, check valve, isolation valve

Filtration and treatment

Maintains coolant cleanliness and chemistry.

Components: coolant filter housing, replaceable filter cartridge, glycol concentration sensor, fill and drain manifold

Expansion and air removal

Accommodates fluid expansion and removes trapped air.

Components: expansion vessel, automatic air vent, pressure relief valve, low pressure switch

Heat exchanger manifold

Transfers rack heat into the outdoor heat rejection path while preserving service bypass options.

Components: plate heat exchanger, temperature balancing valve, purge valve set, service bypass loop, insulated hose set

Operator HMI And Labelling

Presents system status, controls access and communicates hazards to operators.

- site_scada
- service_access
- alarm_bus

Local HMI panel

Provides local status, alarms and manual controls.

Components: industrial touchscreen, panel PC, key switch, alarm buzzer beacon

External labelling

Communicates hazards and service instructions.

Components: high voltage warning label, arc flash boundary label, fire suppression instruction placard, asset QR label

Operator workflow aids

Supports safe inspection and reset actions.

Components: laminated isolation checklist, inspection log holder, status stack light, manual reset station

Remote notification interface

Routes actionable alarms to the site operator while preventing unsafe remote reset.

Components: operator notification relay, SMS alarm output module, alarm acknowledgement button, escalation contact label, remote reset inhibit relay

Bill of Materials

Pricing status: early estimate. Verify supplier, manufacturer, part number, lead time and cost before procurement use.

Total CAPEX BoM: **£41,106**

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
LFP prismatic cells	1	candidate	£100	£100	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-1-lfp_prismatic_cells	8
cell-to-cell busbar	1	candidate	£101	£101	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-2-cell_to_cell_busbar	8
cell terminal hardware set	1	candidate	£102	£102	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-3-cell_terminal_hardware_set	8
cell insulation pad	1	candidate	£103	£103	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-4-cell_insulation_pad	8
module fuse	1	candidate	£104	£104	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-5-module_fuse	8
module steel frame	1	candidate	£105	£105	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-6-module_steel_frame	8
compression plate	1	candidate	£106	£106	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-7-compression_plate	8
compression tie rod set	1	candidate	£107	£107	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-8-compression_tie_rod_set	8
module top cover	1	candidate	£108	£108	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-9-module_top_cover	8
rack label plate	1	candidate	£109	£109	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-10-rack_label_plate	8
BMS slave PCB	1	candidate	£110	£110	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-11-bms_slave_pcb	8
AFE monitor IC	1	candidate	£111	£111	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-12-afe_monitor_ic	8
NTC thermistor	1	candidate	£112	£112	Protocol Evidence	Protocol Evidence	PROTOCOL-13-	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
harness					Supplier	Manufacturer	ntc_thermistor_harness	
isolated CAN transceiver	1	candidate	£113	£113	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-14-isolated_can_transceiver	8
conformal coating	1	candidate	£114	£114	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-15-conformal_coating	8
DC contactor or breaker	1	candidate	£115	£115	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-16-dc_contactor_or_breaker	8
precharge resistor	1	candidate	£116	£116	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-17-precharge_resistor	8
HV fuse holder	1	candidate	£117	£117	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-18-hv_fuse_holder	8
pack current sensor	1	candidate	£118	£118	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-19-pack_current_sensor	8
service disconnect	1	candidate	£119	£119	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-20-service_disconnect	8
cell voltage sense loom	1	candidate	£120	£120	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-21-cell_voltage_sense_loom	8
rack CAN drop cable	1	candidate	£121	£121	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-22-rack_can_drop_cable	8
harness strain relief rail	1	candidate	£122	£122	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-23-harness_strain_relief_rail	8
low-smoke cable sleeve	1	candidate	£123	£123	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-24-low_smoke_cable_sleeve	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
rack cold plate manifold	1	candidate	£124	£124	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-25-rack_cold_plate_manifold	8
thermally conductive gap pad	1	candidate	£125	£125	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-26-thermally_conductive_gap_pad	8
coolant quick connector pair	1	candidate	£126	£126	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-27-coolant_quick_connector_pair	8
rack drip tray	1	candidate	£127	£127	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-28-rack_drip_tray	8
leak detection rope	1	candidate	£128	£128	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-29-leak_detection_rope	8
PCS inverter	1	candidate	£129	£129	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-30-pcs_inverter	8
IGBT power module	1	candidate	£130	£130	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-31-igbt_power_module	8
DC link capacitor bank	1	candidate	£131	£131	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-32-dc_link_capacitor_bank	8
gate driver board	1	candidate	£132	£132	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-33-gate_driver_board	8
inverter control board	1	candidate	£133	£133	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-34-inverter_control_board	8
liquid-cooled heatsink	1	candidate	£134	£134	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-35-liquid_cooled_heatsink	8
cast resin transformer	1	candidate	£135	£135	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-36-cast_resin_transformer	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
HV winding	1	candidate	£136	£136	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-37-hv_winding	8
LV winding	1	candidate	£137	£137	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-38-lv_winding	8
temperature probe	1	candidate	£138	£138	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-39-temperature_probe	8
anti-vibration mount	1	candidate	£139	£139	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-40-anti_vibration_mount	8
grid filter choke	1	candidate	£140	£140	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-41-grid_filter_choke	8
AC filter capacitor	1	candidate	£141	£141	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-42-ac_filter_capacitor	8
EMI filter assembly	1	candidate	£142	£142	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-43-emi_filter_assembly	8
protective earth bar	1	candidate	£143	£143	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-44-protective_earth_bar	8
grid protection relay	1	candidate	£144	£144	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-45-grid_protection_relay	8
voltage sensing fuse set	1	candidate	£145	£145	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-46-voltage_sensing_fuse_set	8
relay test block	1	candidate	£146	£146	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-47-relay_test_block	8
trip circuit monitor	1	candidate	£147	£147	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-48-trip_circuit_monitor	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
G99 settings label	1	candidate	£148	£148	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-49-g99_settings_label	8
corten steel side panel	1	candidate	£149	£149	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-50-corten_steel_side_panel	8
corner casting	1	candidate	£150	£150	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-51-corner_casting	8
marine plywood floor	1	candidate	£151	£151	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-52-marine_plywood_floor	8
roof panel	1	candidate	£152	£152	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-53-roof_panel	8
steel stud frame	1	candidate	£153	£153	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-54-steel_stud_frame	8
fire-rated board	1	candidate	£154	£154	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-55-fire Rated Board	8
mineral wool insulation	1	candidate	£155	£155	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-56-mineral_wool_insulation	8
intumescent sealant	1	candidate	£156	£156	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-57-intumescent_sealant	8
floor reinforcement plate	1	candidate	£157	£157	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-58-floor_reinforcement_plate	8
rack mounting rail	1	candidate	£158	£158	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-59-rack_mounting_rail	8
vibration isolator	1	candidate	£159	£159	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-60-vibration_isolator	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
anchor bolt set	1	candidate	£160	£160	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-61-anchor_bolt_set	8
gland plate assembly	1	candidate	£161	£161	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-62-gland_plate_assembly	8
firestop collar set	1	candidate	£162	£162	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-63-firestop_collar_set	8
EMC cable gland	1	candidate	£163	£163	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-64-emc_cable_gland	8
rain drip shield	1	candidate	£164	£164	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-65-rain_drip_shield	8
penetration schedule label	1	candidate	£165	£165	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-66-penetration_schedule_label	8
DC voltage transducer	1	candidate	£166	£166	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-67-dc_voltage_transducer	8
Hall current sensor	1	candidate	£167	£167	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-68-hall_current_sensor	8
insulation monitoring device	1	candidate	£168	£168	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-69-insulation_monitoring_device	8
auxiliary metering transducer	1	candidate	£169	£169	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-70-auxiliary_metering_transducer	8
temperature humidity sensor	1	candidate	£170	£170	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-71-temperature_humidity_sensor	8
hydrogen detector	1	candidate	£171	£171	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-72-hydrogen_detector	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
smoke aspirating detector	1	candidate	£172	£172	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-73-smoke_aspirating_detector	8
door position switch	1	candidate	£173	£173	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-74-door_position_switch	8
coolant inlet thermistor	1	candidate	£174	£174	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-75-coolant_inlet_thermistor	8
coolant outlet thermistor	1	candidate	£175	£175	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-76-coolant_outlet_thermistor	8
flow meter	1	candidate	£176	£176	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-77-flow_meter	8
pressure transducer	1	candidate	£177	£177	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-78-pressure_transducer	8
fault recorder module	1	candidate	£178	£178	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-79-fault_recorder_module	8
event timestamp logger	1	candidate	£179	£179	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-80-event_timestamp_logger	8
digital input card	1	candidate	£180	£180	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-81-digital_input_card	8
local historian buffer	1	candidate	£181	£181	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-82-local_historian_buffer	8
calibration certificate holder	1	candidate	£182	£182	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-83-calibration_certificate_holder	8
BMS master controller	1	candidate	£183	£183	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-84-bms_master_controller	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
isolation interface	1	candidate	£184	£184	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-85-isolation_interface	8
contactor driver output	1	candidate	£185	£185	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-86-contactor_driver_output	8
SOC estimation firmware	1	candidate	£186	£186	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-87-soc_estimation_firmware	8
industrial controller	1	candidate	£187	£187	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-88-industrial_controller	8
I/O expansion module	1	candidate	£188	£188	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-89-i_o_expansion_module	8
network switch	1	candidate	£189	£189	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-90-network_switch	8
cellular router	1	candidate	£190	£190	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-91-cellular_router	8
Modbus TCP gateway	1	candidate	£191	£191	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-92-modbus_tcp_gateway	8
SCADA protocol adapter	1	candidate	£192	£192	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-93-scada_protocol_adapter	8
time synchronisation module	1	candidate	£193	£193	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-94-time_synchronisation_module	8
cybersecurity firewall	1	candidate	£194	£194	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-95-cybersecurity_firewall	8
VPN access appliance	1	candidate	£195	£195	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-96-vpn_access_appliance	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
syslog collector	1	candidate	£196	£196	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-97-syslog_collector	8
account key switch	1	candidate	£197	£197	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-98-account_key_switch	8
firmware escrow tag	1	candidate	£198	£198	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-99-firmware_escrow_tag	8
configuration backup module	1	candidate	£199	£199	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-100-configuration_backup_module	8
Fire detection/suppression	1	candidate	£200	£200	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-101-fire_detection_suppression	8
clean-agent cylinder	1	candidate	£201	£201	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-102-clean_agent_cylinder	8
release solenoid	1	candidate	£202	£202	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-103-release_solenoid	8
nozzle pipework	1	candidate	£203	£203	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-104-nozzle_pipework	8
pressure switch	1	candidate	£204	£204	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-105-pressure_switch	8
E-stop mushroom button	1	candidate	£205	£205	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-106-e_stop_mushroom_button	8
safety relay	1	candidate	£206	£206	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-107-safety_relay	8
DC shunt trip output	1	candidate	£207	£207	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-108-dc_shunt_trip_output	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
AC breaker trip coil	1	candidate	£208	£208	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-109-ac_breaker_trip_coil	8
DC surge protection device	1	candidate	£209	£209	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-110-dc_surge_protection_device	8
AC surge protection device	1	candidate	£210	£210	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-111-ac_surge_protection_device	8
arc flash label set	1	candidate	£211	£211	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-112-arc_flash_label_set	8
protective earth bonding kit	1	candidate	£212	£212	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-113-protective_earth_bonding_kit	8
pressure relief vent panel	1	candidate	£213	£213	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-114-pressure_relief_vent_panel	8
vent microswitch	1	candidate	£214	£214	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-115-vent_microswitch	8
flame arrestor mesh	1	candidate	£215	£215	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-116-flame_arrestor_mesh	8
vent weather hood	1	candidate	£216	£216	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-117-vent_weather_hood	8
post-event inspection tag	1	candidate	£217	£217	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-118-post_event_inspection_tag	8
container HVAC unit	1	candidate	£218	£218	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-119-container_hvac_unit	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
air filter element	1	candidate	£219	£219	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-120-air_filter_element	8
condensate drain kit	1	candidate	£220	£220	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-121-condensate_drain_kit	8
HVAC controller	1	candidate	£221	£221	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-122-hvac_controller	8
intake louvre	1	candidate	£222	£222	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-123-intake_louvre	8
exhaust louvre	1	candidate	£223	£223	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-124-exhaust_louvre	8
EC fan assembly	1	candidate	£224	£224	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-125-ec_fan_assembly	8
airflow baffle	1	candidate	£225	£225	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-126-airflow_baffle	8
insulation panel	1	candidate	£226	£226	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-127-insulation_panel	8
vapour barrier film	1	candidate	£227	£227	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-128-vapour_barrier_film	8
thermal break strip	1	candidate	£228	£228	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-129-thermal_break_strip	8
seal gasket set	1	candidate	£229	£229	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-130-seal_gasket_set	8
dry cooler coil	1	candidate	£230	£230	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-131-dry_cooler_coil	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
EC condenser fan	1	candidate	£231	£231	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-132-ec_condenser_fan	8
fan contactor	1	candidate	£232	£232	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-133-fan_contactor	8
coil guard grille	1	candidate	£233	£233	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-134-coil_guard_grille	8
drain-safe condensate pan	1	candidate	£234	£234	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-135-drain_safe_condensate_pan	8
copper DC busbar	1	candidate	£235	£235	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-136-copper_dc_busbar	8
busbar support insulator	1	candidate	£236	£236	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-137-busbar_support_insulator	8
busbar shroud	1	candidate	£237	£237	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-138-busbar_shroud	8
torque witness marker	1	candidate	£238	£238	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-139-torque_witness_marker	8
moulded-case circuit breaker	1	candidate	£239	£239	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-140-moulded_case_circuit_breaker	8
AC contactor	1	candidate	£240	£240	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-141-ac_contactor	8
metering CT set	1	candidate	£241	£241	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-142-metering_ct_set	8
terminal block assembly	1	candidate	£242	£242	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-143-terminal_block_assembly	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
auxiliary transformer	1	candidate	£243	£243	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-144-auxiliary_transformer	8
24 V DC power supply	1	candidate	£244	£244	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-145-24_v_dc_power_supply	8
miniature circuit breaker bank	1	candidate	£245	£245	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-146-miniature_circuit_breaker_bank	8
UPS module	1	candidate	£246	£246	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-147-ups_module	8
main earth bar	1	candidate	£247	£247	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-148-main_earth_bar	8
earth stud array	1	candidate	£248	£248	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-149-earth_stud_array	8
lightning surge counter	1	candidate	£249	£249	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-150-lightning_surge_counter	8
equipotential bonding strap	1	candidate	£250	£250	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-151-equipotential_bonding_strap	8
earth continuity test point	1	candidate	£251	£251	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-152-earth_continuity_test_point	8
lockable access door	1	candidate	£252	£252	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-153-lockable_access_door	8
door seal gasket	1	candidate	£253	£253	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-154-door_seal_gasket	8
panic release latch	1	candidate	£254	£254	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-155-panic_release_latch	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
door stay arm	1	candidate	£255	£255	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-156-door_stay_arm	8
LED service luminaire	1	candidate	£256	£256	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-157-led_service_luminaire	8
emergency light fitting	1	candidate	£257	£257	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-158-emergency_light_fitting	8
maintenance receptacle	1	candidate	£258	£258	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-159-maintenance_receptacle	8
lighting switchgear	1	candidate	£259	£259	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-160-lighting_switchgear	8
certified lifting lug	1	candidate	£260	£260	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-161-certified_lifting_lug	8
forklift pocket reinforcement	1	candidate	£261	£261	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-162-forklift_pocket_reinforcement	8
removable rack guide	1	candidate	£262	£262	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-163-removable_rack_guide	8
service trolley interface	1	candidate	£263	£263	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-164-service_trolley_interface	8
insulated test socket	1	candidate	£264	£264	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-165-insulated_test_socket	8
coolant sampling port	1	candidate	£265	£265	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-166-coolant_sampling_port	8
laptop service port	1	candidate	£266	£266	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-167-laptop_service_port	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
lockout hasp rail	1	candidate	£267	£267	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 168- lockout_has p_rail	8
commissioning torque log plate	1	candidate	£268	£268	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 169- commissioning_torque_log_plate	8
Thermal management loop	1	candidate	£269	£269	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 170- thermal_management_loop	8
coolant circulation pump	1	candidate	£270	£270	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 171- coolant_circulation_pump	8
pump VFD	1	candidate	£271	£271	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 172- pump_vfd	8
check valve	1	candidate	£272	£272	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 173- check_valve	8
isolation valve	1	candidate	£273	£273	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 174- isolation_valve	8
coolant filter housing	1	candidate	£274	£274	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 175- coolant_filter_housing	8
replaceable filter cartridge	1	candidate	£275	£275	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 176- replaceable_filter_cartridge	8
glycol concentration sensor	1	candidate	£276	£276	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 177- glycol_concentration_sensor	8
fill and drain manifold	1	candidate	£277	£277	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL- 178- fill_and_drain_manifold	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
expansion vessel	1	candidate	£278	£278	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-179-expansion_vessel	8
automatic air vent	1	candidate	£279	£279	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-180-automatic_air_vent	8
pressure relief valve	1	candidate	£280	£280	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-181-pressure_relief_valve	8
low pressure switch	1	candidate	£281	£281	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-182-low_pressure_switch	8
plate heat exchanger	1	candidate	£282	£282	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-183-plate_heat_exchanger	8
temperature balancing valve	1	candidate	£283	£283	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-184-temperature_balancing_valve	8
purge valve set	1	candidate	£284	£284	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-185-purge_valve_set	8
service bypass loop	1	candidate	£285	£285	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-186-service_bypass_loop	8
insulated hose set	1	candidate	£286	£286	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-187-insulated_hose_set	8
industrial touchscreen	1	candidate	£287	£287	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-188-industrial_touchscreen	8
panel PC	1	candidate	£288	£288	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-189-panel_pc	8
key switch	1	candidate	£289	£289	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-190-key_switch	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
alarm buzzer beacon	1	candidate	£290	£290	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-191-alarm_buzzer_beacon	8
high voltage warning label	1	candidate	£291	£291	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-192-high_voltage_warning_label	8
arc flash boundary label	1	candidate	£292	£292	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-193-arc_flash_boundary_label	8
fire suppression instruction placard	1	candidate	£293	£293	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-194-fire_suppression_instruction_placard	8
asset QR label	1	candidate	£294	£294	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-195-asset_qr_label	8
laminated isolation checklist	1	candidate	£295	£295	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-196-laminated_isolation_checklist	8
inspection log holder	1	candidate	£296	£296	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-197-inspection_log_holder	8
status stack light	1	candidate	£297	£297	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-198-status_stack_light	8
manual reset station	1	candidate	£298	£298	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-199-manual_reset_station	8
operator notification relay	1	candidate	£299	£299	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-200-operator_notification_relay	8
SMS alarm output module	1	candidate	£300	£300	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-201-sms_alarm_output_module	8
alarm acknowledged	1	candidate	£301	£301	Protocol Evidence	Protocol Evidence	PROTOCOL-202-	8

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
ement button					Supplier	Manufactur er	alarm_ackn owledgeme nt_button	
escalation contact label	1	candidate	£302	£302	Protocol Evidence Supplier	Protocol Evidence Manufactur er	PROTOCOL- 203- escalation_ contact_lab el	8
remote reset inhibit relay	1	candidate	£303	£303	Protocol Evidence Supplier	Protocol Evidence Manufactur er	PROTOCOL- 204- remote_res et_inhibit_r elay	8

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
complete	204	204	0	0

Critical Sourcing Worklist

No critical sourcing blockers.

No rejected source records.

BoM Sourcing Ledger

204/204 priced lines admitted. Critical coverage 6/6. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
LFP prismatic cells	admitted	unique	critical	£100	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-1-lfp-prismatic_cells	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	Keep source record attached; re-check freshness before procurement use.
cell-to-cell busbar	admitted	unique	candidate	£101	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-2-cell_to_cell_busbar	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	Keep source record attached; re-check freshness before procurement use.
cell terminal hardware set	admitted	unique	candidate	£102	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-3-cell_terminal_hardware_set	test-fixture://review-evidence-closure/source/cell_terminal_hardware_set	Keep source record attached; re-check freshness before procurement use.
cell insulation pad	admitted	unique	candidate	£103	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-4-cell_insulation_pad	test-fixture://review-evidence-closure/source/cell_insulation_pad	Keep source record attached; re-check freshness before procurement use.
module fuse	admitted	unique	candidate	£104	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-5-module_fuse	test-fixture://review-evidence-closure/source/module_fuse	Keep source record attached; re-check freshness before procurement use.
module steel frame	admitted	unique	candidate	£105	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-6-module_steel_frame	test-fixture://review-evidence-closure/source/module_steel_frame	Keep source record attached; re-check freshness before procurement use.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
compression plate	admitted	unique	candidate	£106	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-7-compression_plate	test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; re-check freshness before procurement use.
compression tie rod set	admitted	unique	candidate	£107	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-8-compression_tie_rod_set	test-fixture://review-evidence-closure/source/compression_tie_rod_set	Keep source record attached; re-check freshness before procurement use.
module top cover	admitted	unique	candidate	£108	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-9-module_top_cover	test-fixture://review-evidence-closure/source/module_top_cover	Keep source record attached; re-check freshness before procurement use.
rack label plate	admitted	unique	candidate	£109	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-10-rack_label_plate	test-fixture://review-evidence-closure/source/rack_label_plate	Keep source record attached; re-check freshness before procurement use.
BMS slave PCB	admitted	unique	candidate	£110	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-11-bms_slave_pcb	test-fixture://review-evidence-closure/source/bms_slave_pcb	Keep source record attached; re-check freshness before procurement use.
AFE monitor IC	admitted	unique	candidate	£111	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-12-afe_monitor_ic	test-fixture://review-evidence-closure/source/afe_monitor_ic	Keep source record attached; re-check freshness before procurement use.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
NTC thermistor harness	admitted	unique	candidate	£112	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-13-ntc_thermistor_harness	test-fixture://review-evidence-closure/source/ntc_thermistor_harness	Keep source record attached; re-check freshness before procurement use.
isolated CAN transceiver	admitted	unique	candidate	£113	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-14-isolated_can_transceiver	test-fixture://review-evidence-closure/source/isolated_can_transceiver	Keep source record attached; re-check freshness before procurement use.
conformal coating	admitted	unique	candidate	£114	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-15-conformal_coating	test-fixture://review-evidence-closure/source/conformal_coating	Keep source record attached; re-check freshness before procurement use.
DC contactor or breaker	admitted	unique	critical	£115	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-16-dc_contactor_or_breaker	test-fixture://review-evidence-closure/source/dc_contactor_or_breaker	Keep source record attached; re-check freshness before procurement use.
precharge resistor	admitted	unique	candidate	£116	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-17-precharge_resistor	test-fixture://review-evidence-closure/source/precharge_resistor	Keep source record attached; re-check freshness before procurement use.
HV fuse holder	admitted	unique	candidate	£117	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-18-hv_fuse_holder	test-fixture://review-evidence-closure/source/hv_fuse_holder	Keep source record attached; re-check freshness before procurement use.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
pack current sensor	admitted	unique	candidate	£118	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-19-pack_current_sensor	test-fixture://review-evidence-closure/source/pack_current_sensor	Keep source record attached; re-check freshness before procurement use.
service disconnect	admitted	unique	candidate	£119	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-20-service_disconnect	test-fixture://review-evidence-closure/source/service_disconnect	Keep source record attached; re-check freshness before procurement use.
cell voltage sense loom	admitted	unique	candidate	£120	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-21-cell_voltage_sense_loom	test-fixture://review-evidence-closure/source/cell_voltage_sense_loom	Keep source record attached; re-check freshness before procurement use.
rack CAN drop cable	admitted	unique	candidate	£121	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-22-rack_can_drop_cable	test-fixture://review-evidence-closure/source/rack_can_drop_cable	Keep source record attached; re-check freshness before procurement use.
harness strain relief rail	admitted	unique	candidate	£122	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-23-harness_strain_relief_rail	test-fixture://review-evidence-closure/source/harness_strain_relief_rail	Keep source record attached; re-check freshness before procurement use.
low-smoke cable sleeve	admitted	unique	candidate	£123	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-24-low_smoke_cable_sleeve	test-fixture://review-evidence-closure/source/low_smoke_cable_sleeve	Keep source record attached; re-check freshness before procurement use.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
rack cold plate manifold	admitted	unique	candidate	£124	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-25-rack_cold_plate_manifold	test-fixture://review-evidence-closure/source/rack_cold_plate_manifold	Keep source record attached; re-check freshness before procurement use.
thermally conductive gap pad	admitted	unique	candidate	£125	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-26-thermally_conductive_gap_pad	test-fixture://review-evidence-closure/source/thermally_conductive_gap_pad	Keep source record attached; re-check freshness before procurement use.
coolant quick connector pair	admitted	unique	candidate	£126	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-27-coolant_quick_connector_pair	test-fixture://review-evidence-closure/source/coolant_quick_connector_pair	Keep source record attached; re-check freshness before procurement use.
rack drip tray	admitted	unique	candidate	£127	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-28-rack_drip_tray	test-fixture://review-evidence-closure/source/rack_drip_tray	Keep source record attached; re-check freshness before procurement use.
leak detection rope	admitted	unique	candidate	£128	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-29-leak_detection_rope	test-fixture://review-evidence-closure/source/leak_detection_rope	Keep source record attached; re-check freshness before procurement use.
PCS inverter	admitted	unique	critical	£129	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-30-pcs_inverter	test-fixture://review-evidence-closure/source/pcs_inverter	Keep source record attached; re-check freshness before procurement use.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
IGBT power module	admitted	unique	candidate	£130	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-31-igbt_power_module	test-fixture://review-evidence-closure/source/igbt_power_module	Keep source record attached; re-check freshness before procurement use.
DC link capacitor bank	admitted	unique	candidate	£131	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-32-dc_link_capacitor_bank	test-fixture://review-evidence-closure/source/dc_link_capacitor_bank	Keep source record attached; re-check freshness before procurement use.
gate driver board	admitted	unique	candidate	£132	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-33-gate_driver_board	test-fixture://review-evidence-closure/source/gate_driver_board	Keep source record attached; re-check freshness before procurement use.
inverter control board	admitted	unique	candidate	£133	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-34-inverter_control_board	test-fixture://review-evidence-closure/source/inverter_control_board	Keep source record attached; re-check freshness before procurement use.
liquid-cooled heatsink	admitted	unique	candidate	£134	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-35-liquid_cooled_heatsink	test-fixture://review-evidence-closure/source/liquid_cooled_heatsink	Keep source record attached; re-check freshness before procurement use.
cast resin transformer	admitted	unique	candidate	£135	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-36-cast_resin_transformer	test-fixture://review-evidence-closure/source/cast_resin_transformer	Keep source record attached; re-check freshness before procurement use.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
HV winding	admitted	unique	candidate	£136	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-37-hv_winding	test-fixture://review-evidence-closure/source/hv_winding	Keep source record attached; re-check freshness before procurement use.
LV winding	admitted	unique	candidate	£137	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-38-lv_winding	test-fixture://review-evidence-closure/source/lv_winding	Keep source record attached; re-check freshness before procurement use.
temperature probe	admitted	unique	candidate	£138	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-39-temperature_probe	test-fixture://review-evidence-closure/source/temperature_probe	Keep source record attached; re-check freshness before procurement use.
anti-vibration mount	admitted	unique	candidate	£139	Protocol Evidence Supplier	Protocol Evidence Manufacturer	PROTOCOL-40-anti_vibration_mount	test-fixture://review-evidence-closure/source/anti_vibration_mount	Keep source record attached; re-check freshness before procurement use.

Showing first 40 of 204 ledger rows; export CSV for the full line-by-line ledger.

BoM Provenance Manifest

1020 source-backed claims, 0 missing-source claims, 0 provenance violations. Critical missing-source claims: 0. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
LFP prismatic cells	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	Keep source record attached; refresh evidence before procurement use.
LFP prismatic cells	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	Keep source record attached; refresh evidence before procurement use.
LFP prismatic cells	mpn	source backed	PROTOCOL-1-lfp_prismatic_cells	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	Keep source record attached; refresh evidence before procurement use.
LFP prismatic cells	unit_cost_gbp	source backed	100	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	Keep source record attached; refresh evidence before procurement use.
LFP prismatic cells	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/lfp_prismatic_cells	Keep source record attached; refresh evidence before procurement use.
cell-to-cell busbar	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	Keep source record attached; refresh evidence before procurement use.
cell-to-cell busbar	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	Keep source record attached; refresh evidence before procurement use.
cell-to-cell busbar	mpn	source backed	PROTOCOL-2-cell_to_cell_busbar	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	Keep source record attached; refresh evidence before procurement use.
cell-to-cell busbar	unit_cost_gbp	source backed	101	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	Keep source record attached; refresh evidence before procurement use.
cell-to-cell busbar	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/cell_to_cell_busbar	Keep source record attached; refresh evidence before procurement use.
cell terminal hardware set	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/cell_	Keep source record attached; refresh evidence before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
cell terminal hardware set	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-evidence-closure/source/cell_terminal_hardware_set	Keep source record attached; refresh evidence before procurement use.
	mpn	source backed	PROTOCOL-3-cell_terminal_hardware_set	test-fixture://review-evidence-closure/source/cell_terminal_hardware_set	Keep source record attached; refresh evidence before procurement use.
	unit_cost_gbp	source backed	102	test-fixture://review-evidence-closure/source/cell_terminal_hardware_set	Keep source record attached; refresh evidence before procurement use.
	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/cell_terminal_hardware_set	Keep source record attached; refresh evidence before procurement use.
cell insulation pad	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/cell_insulation_pad	Keep source record attached; refresh evidence before procurement use.
cell insulation pad	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-evidence-closure/source/cell_insulation_pad	Keep source record attached; refresh evidence before procurement use.
cell insulation pad	mpn	source backed	PROTOCOL-4-cell_insulation_pad	test-fixture://review-evidence-closure/source/cell_insulation_pad	Keep source record attached; refresh evidence before procurement use.
cell insulation pad	unit_cost_gbp	source backed	103	test-fixture://review-evidence-closure/source/cell_insulation_pad	Keep source record attached; refresh evidence before procurement use.
cell insulation pad	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/cell_insulation_pad	Keep source record attached; refresh evidence before procurement use.
module fuse	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-	Keep source record attached; refresh evidence before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
module fuse	manufacturer	source backed	Protocol Evidence Manufacturer	closure/source/module_fuse	
				test-fixture://review-evidence-closure/source/module_fuse	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_fuse	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_fuse	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_fuse	Keep source record attached; refresh evidence before procurement use.
module steel frame	supplier	source backed	Protocol Evidence Supplier	closure/source/module_steel_frame	
				test-fixture://review-evidence-closure/source/module_steel_frame	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_steel_frame	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_steel_frame	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_steel_frame	Keep source record attached; refresh evidence before procurement use.
compression plate	manufacturer	source backed	Protocol Evidence Manufacturer	closure/source/compression_plate	
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
compression plate	mpn	source backed	PROTOCOL-7-compression_plate	evidence-closure/source/compression_plate	evidence before procurement use.
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
compression plate	unit_cost_gbp	source backed	106	test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
compression plate	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/compression_plate	Keep source record attached; refresh evidence before procurement use.
compression tie rod set	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/compression_tie_rod_set	Keep source record attached; refresh evidence before procurement use.
compression tie rod set	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-evidence-closure/source/compression_tie_rod_set	Keep source record attached; refresh evidence before procurement use.
compression tie rod set	mpn	source backed	PROTOCOL-8-compression_tie_rod_set	test-fixture://review-evidence-closure/source/compression_tie_rod_set	Keep source record attached; refresh evidence before procurement use.
compression tie rod set	unit_cost_gbp	source backed	107	test-fixture://review-evidence-closure/source/compression_tie_rod_set	Keep source record attached; refresh evidence before procurement use.
compression tie rod set	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/compression_tie_rod_set	Keep source record attached; refresh evidence before procurement use.
module top cover	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/module_top_cover	Keep source record attached; refresh evidence before procurement use.
module top cover	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-	Keep source record attached; refresh

Line	Claim Field	Status	Value	Source Ref	Next Action
module top cover	mpn	source backed	PROTOCOL-9-module_top_cover	evidence-closure/source/module_top_cover	evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_top_cover	Keep source record attached; refresh evidence before procurement use.
				test-fixture://review-evidence-closure/source/module_top_cover	Keep source record attached; refresh evidence before procurement use.
module top cover	unit_cost_gbp	source backed	108	test-fixture://review-evidence-closure/source/module_top_cover	Keep source record attached; refresh evidence before procurement use.
module top cover	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/module_top_cover	Keep source record attached; refresh evidence before procurement use.
rack label plate	supplier	source backed	Protocol Evidence Supplier	test-fixture://review-evidence-closure/source/rack_label_plate	Keep source record attached; refresh evidence before procurement use.
rack label plate	manufacturer	source backed	Protocol Evidence Manufacturer	test-fixture://review-evidence-closure/source/rack_label_plate	Keep source record attached; refresh evidence before procurement use.
rack label plate	mpn	source backed	PROTOCOL-10-rack_label_plate	test-fixture://review-evidence-closure/source/rack_label_plate	Keep source record attached; refresh evidence before procurement use.
rack label plate	unit_cost_gbp	source backed	109	test-fixture://review-evidence-closure/source/rack_label_plate	Keep source record attached; refresh evidence before procurement use.
rack label plate	lead_time_weeks	source backed	8	test-fixture://review-evidence-closure/source/rack_label_plate	Keep source record attached; refresh evidence before procurement use.

Sourcing Evidence Pack

No critical evidence packets remain open.

Cost Bucket	Estimate
CAPEX	£41,106
Annual OPEX	£1,644
NRE	£25,000

Supplier Path	Spend	Risk
Protocol Evidence Supplier	£41,106	medium

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
IEC-62619	Industrial lithium battery safety	Global / UK-adopted	Cell certificates and pack-level safety validation.
UL-9540A	Thermal runaway fire propagation test	US / investor diligence	Cell, module and unit-level propagation evidence.
G99	UK grid connection requirements	UK	Protection settings, relay test certificates and commissioning records.

Hazard	S	O	D	RPN	Mitigation
Thermal runaway propagation	10	3	4	120	Cell spacing, thermal monitoring and clean-agent suppression.
DC arc fault	9	3	4	108	Arc detection, contactor isolation and maintenance access controls.

Section Issues And Score

No section issues found by current deterministic validators.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 9. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	9						